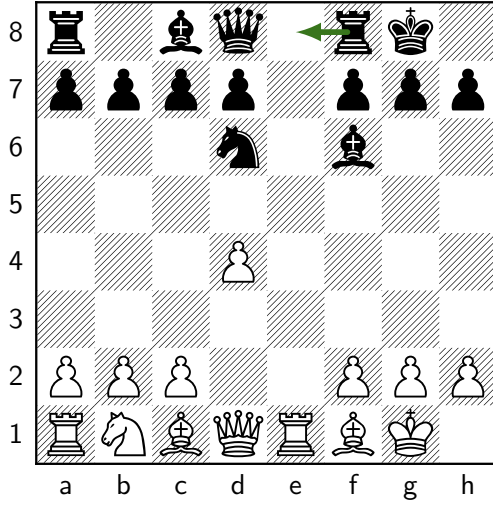


Melanie-MASTERS (corpus: human masters) — match game 4

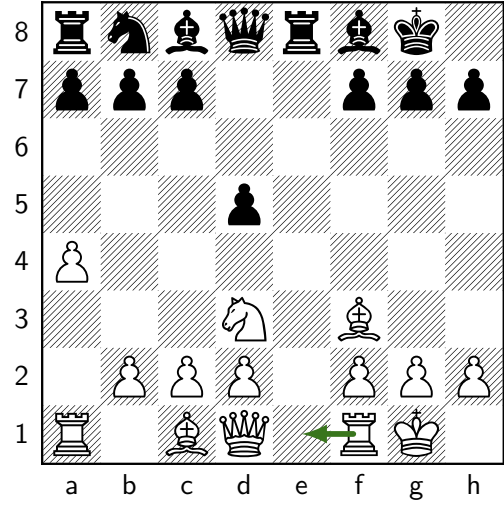
Melanie (Deep_Vision)

Abstract

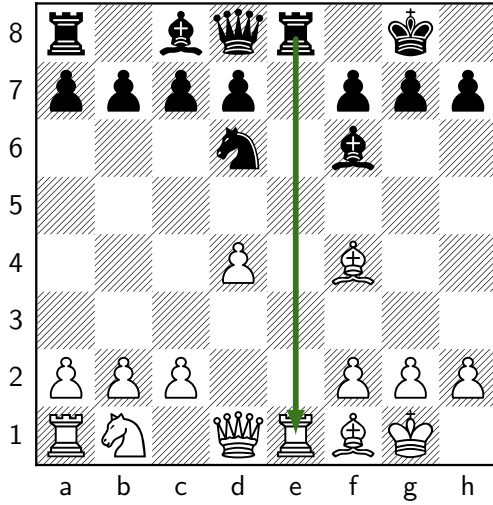
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



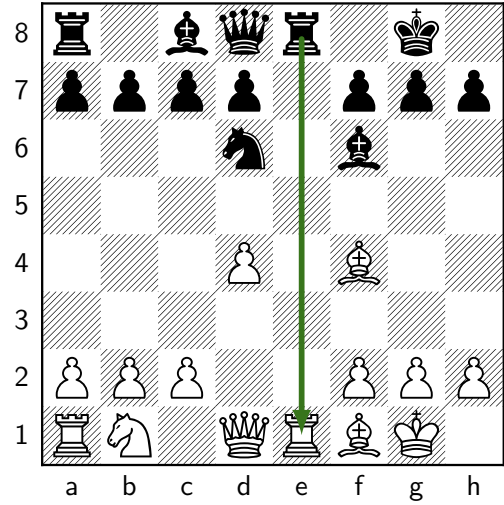
(a)



(b)

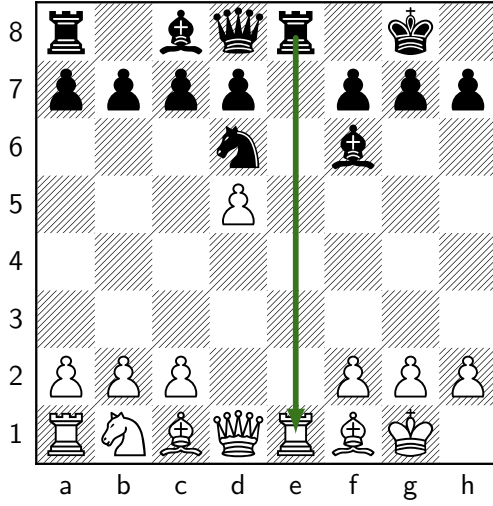


(c)

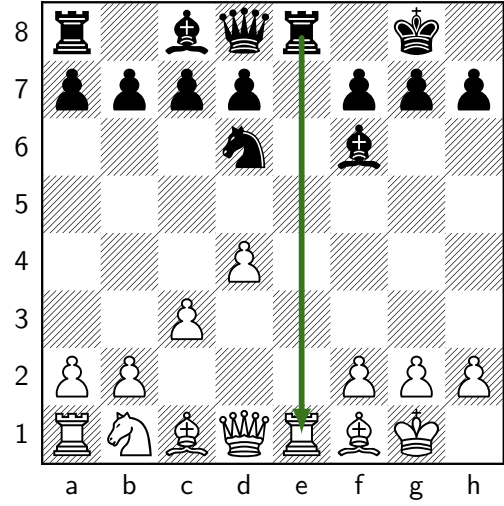


(d)

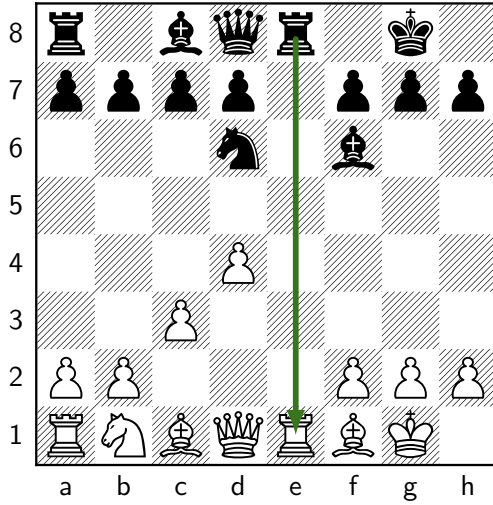
Figure 1: **Move 10, Black: Re8.** (a) Our position; Melanie plays **Re8** (**f8e8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#2448` (ply 20) [structural distance $d=1$, QAP obj=28]: the master played **Re1**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Nd6* as their *Ra1*, our *Ra8* as their *Qd1*; transferred move **f8e8** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#2940` (ply 21) [structural distance $d=32$, QAP obj=13]: the master played **Rxe1**; under μ our *Rf1* plays the role of that game’s *Re1*; cross-type roles: our *Pg7* as their *Ra8*, our *Ph7* as their *Bc8*; transferred move **f8e8** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0005#3526` (ply 21) [structural distance $d=32$, QAP obj=13]: the master played **Rxe1**; under μ our *Rf1* plays the role of that game’s *Re1*; cross-type roles: our *Pg7* as their *Ra8*, our *Ph7* as their *Bc8*; transferred move **f8e8** (**endorses this move**). Synthesis: 23 retrieved analogs transferred to **Re8** (summed QAP objective 307, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



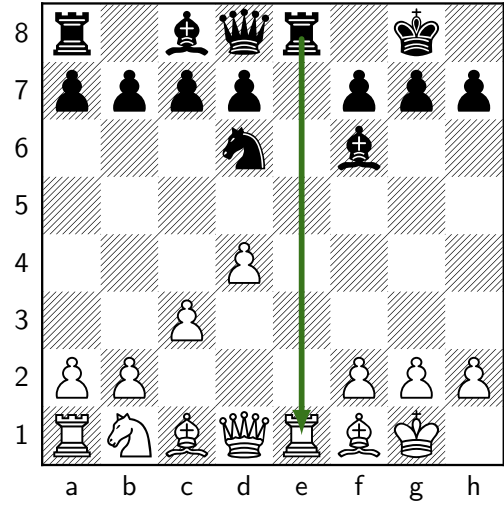
(a)



(b)

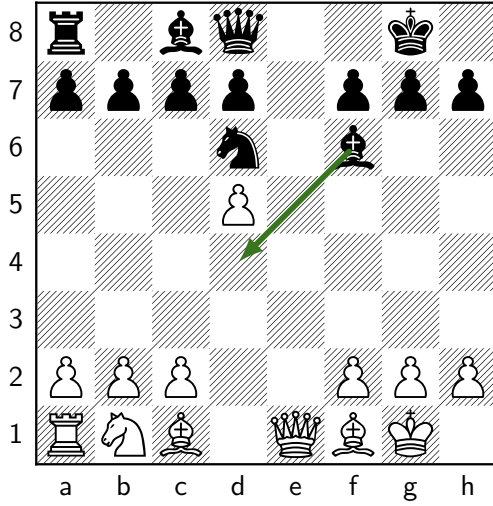


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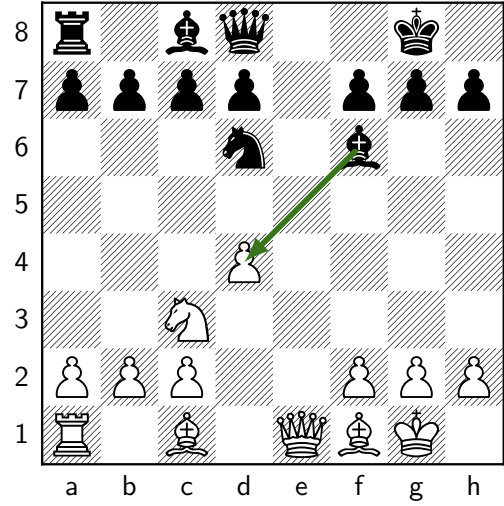


(d)

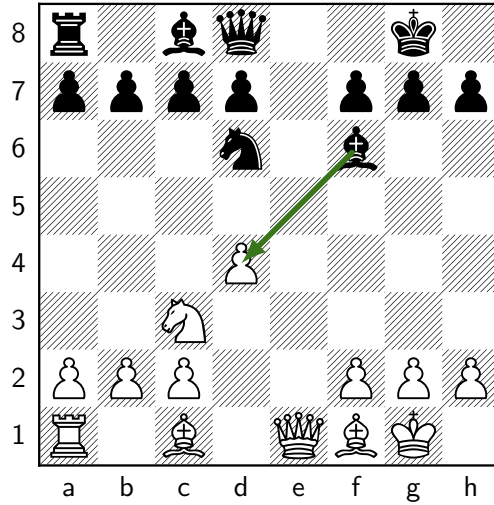
Figure 2: **Move 11, Black: Rxe1**. (a) Our position; Melanie plays Rxe1 (e8e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#389` (ply 21) [structural distance $d=21$, QAP obj=64]: the master played Rxe1; under μ our *Re1* plays the role of that game’s *Re1*; cross-type roles: our *Pg7* as their *Ra8*, our *Bf6* as their *Pb7*; transferred move e8e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#512` (ply 21) [structural distance $d=21$, QAP obj=64]: the master played Rxe1; under μ our *Re1* plays the role of that game’s *Re1*; cross-type roles: our *Pg7* as their *Ra8*, our *Bf6* as their *Pb7*; transferred move e8e1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#1555` (ply 21) [structural distance $d=21$, QAP obj=64]: the master played Rxe1; under μ our *Re1* plays the role of that game’s *Re1*; cross-type roles: our *Pg7* as their *Ra8*, our *Bf6* as their *Pb7*; transferred move e8e1 (**endorses this move**). Synthesis: 64 retrieved analogs transferred to Rxe1 (summed QAP objective 3396, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



(a)

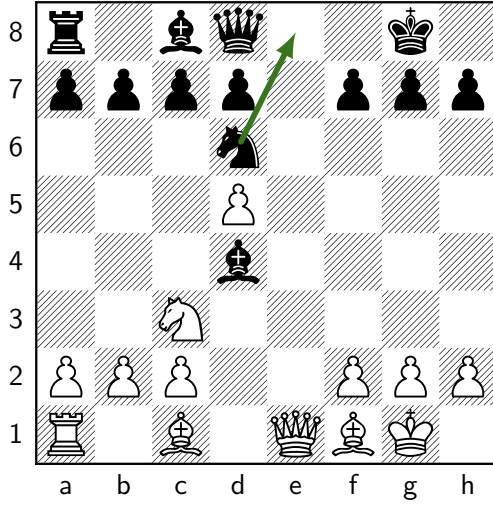


(b)

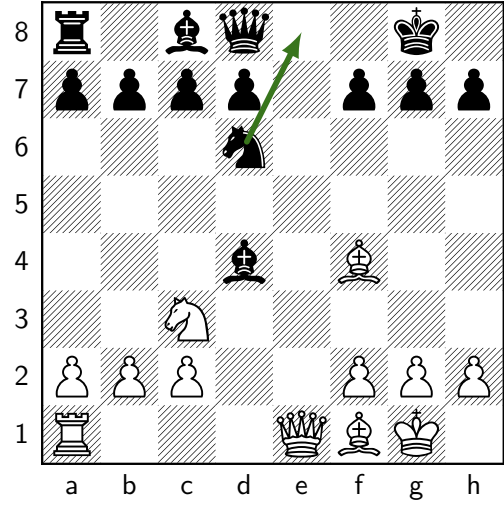


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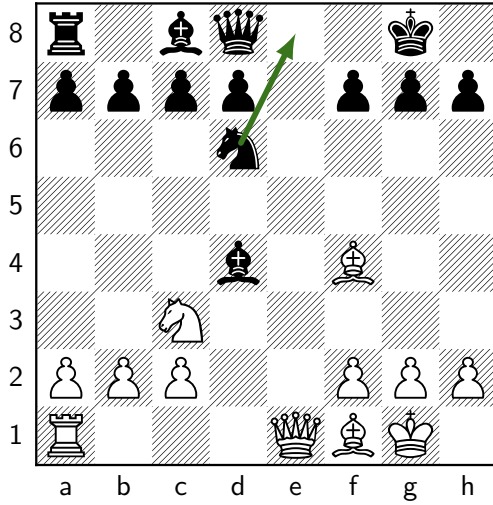
Figure 3: **Move 12, Black: Bd4.** (a) Our position; Melanie plays Bd4 (f6d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#2436` (ply 23) [structural distance $d=24$, QAP obj=33]: the master played Bxd4; under μ our Bf3 plays the role of that game’s Bf3; cross-type roles: our bf1 as their Ra8, our pc2 as their Bc8; transferred move f6d4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#1605` (ply 23) [structural distance $d=24$, QAP obj=33]: the master played Bxd4; under μ our Bf3 plays the role of that game’s Bf3; cross-type roles: our bf1 as their Ra8, our pc2 as their Bc8; transferred move f6d4 (**endorses this move**). Synthesis: 2 retrieved analogs transferred to Bd4 (summed QAP objective 66, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



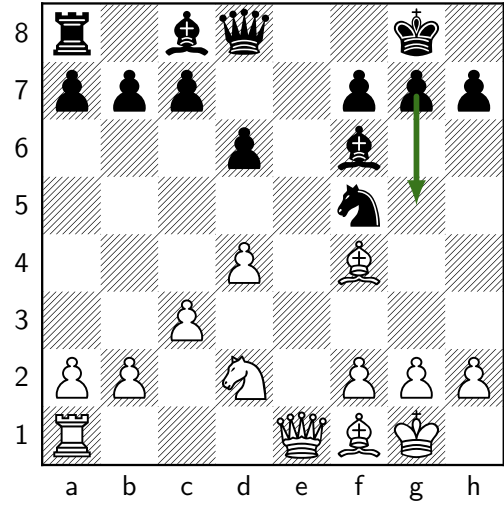
(a)



(b)

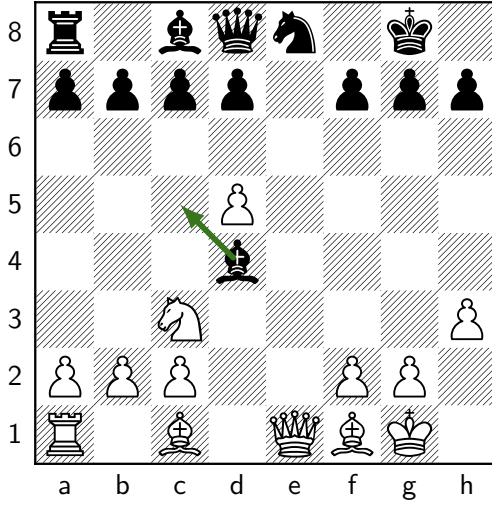


(c)

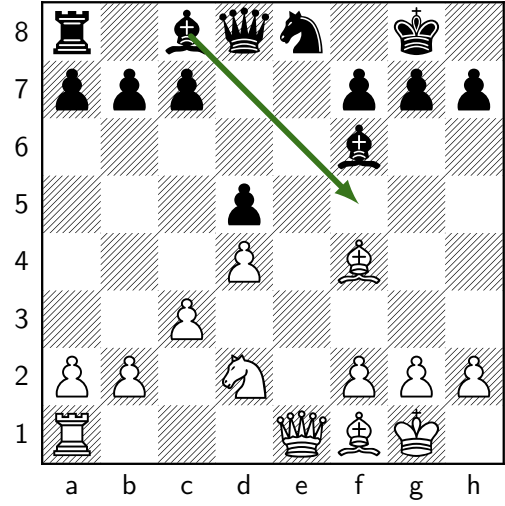


(d)

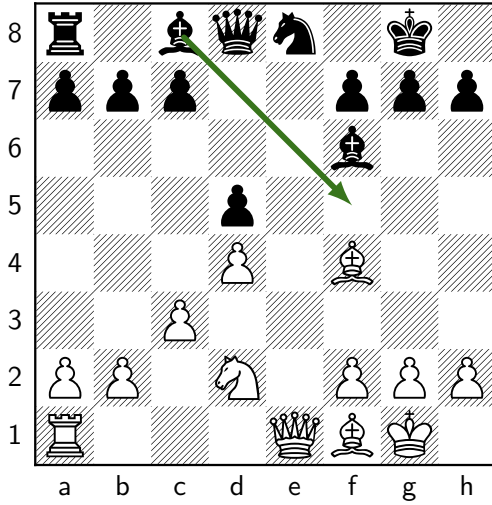
Figure 4: **Move 13, Black: Ne8.** (a) Our position; Melanie plays Ne8 (d6e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#2436` (ply 25) [structural distance $d=31$, QAP obj=81]: the master played Ne8; under μ our Nd3 plays the role of that game’s Nd3; cross-type roles: our pd5 as their bf4; transferred move d6e8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#1605` (ply 25) [structural distance $d=31$, QAP obj=81]: the master played Ne8; under μ our Nd3 plays the role of that game’s Nd3; cross-type roles: our pd5 as their bf4; transferred move d6e8 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#403` (ply 27) [structural distance $d=39$, QAP obj=6]: the master played g5; under μ our Ph2 plays the role of that game’s Pg2; cross-type roles: our Pc7 as their Ra8, our Nd6 as their Pa7; transferred move h7h6. Synthesis: 2 retrieved analogs transferred to Ne8 (summed QAP objective 161, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



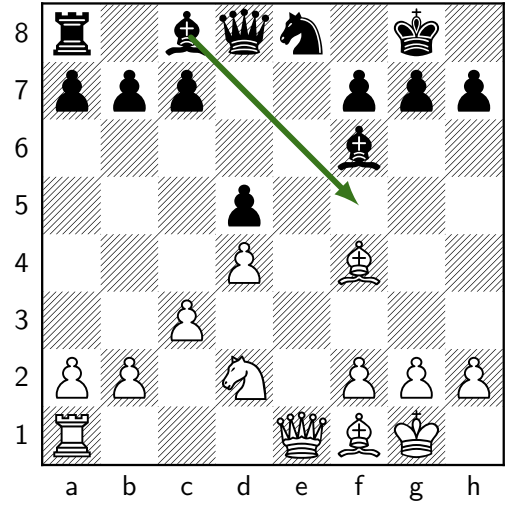
(a)



(b)

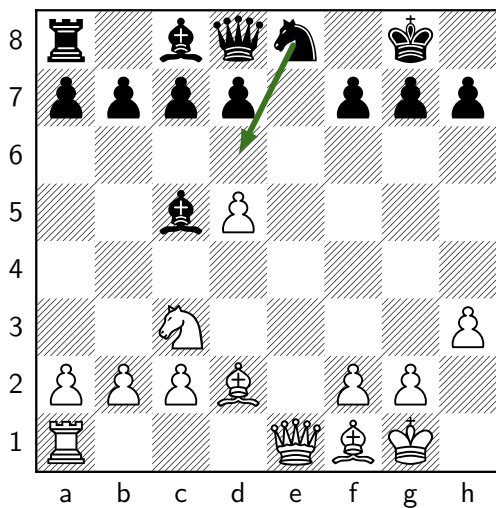


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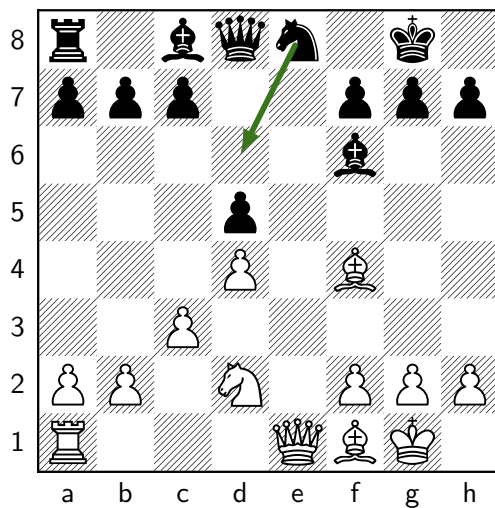


(d)

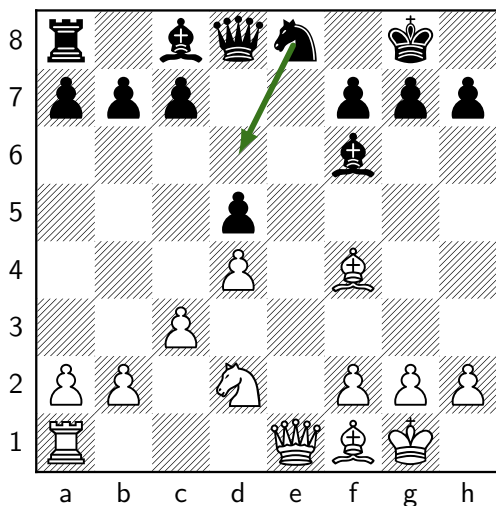
Figure 5: **Move 14, Black: Bc5.** (a) Our position; Melanie plays Bc5 (d4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0012#3688` (ply 27) [structural distance $d=65$, QAP obj=13]: the master played Bf5; under μ our Bd5 plays the role of that game’s Bc1; cross-type roles: our Bc8 as their Pb7, our pd5 as their Bf6; transferred move d4c5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#327` (ply 27) [structural distance $d=65$, QAP obj=13]: the master played Bf5; under μ our Bd5 plays the role of that game’s Bc1; cross-type roles: our Bc8 as their Pb7, our pd5 as their Bf6; transferred move d4c5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0016#690` (ply 27) [structural distance $d=65$, QAP obj=13]: the master played Bf5; under μ our Bd5 plays the role of that game’s Bc1; cross-type roles: our Bc8 as their Pb7, our pd5 as their Bf6; transferred move d4c5 (**endorses this move**). Synthesis: 54 retrieved analogs transferred to Bc5 (summed QAP objective 696, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



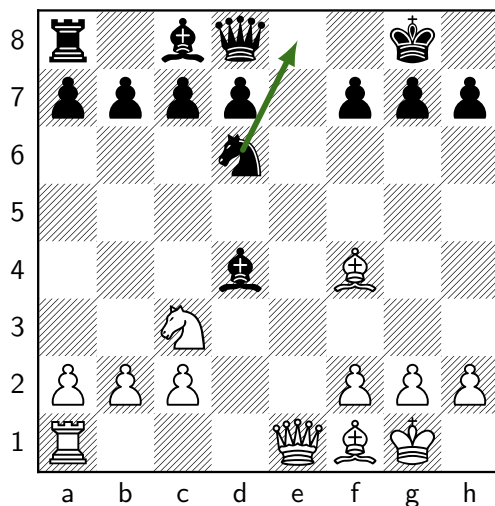
(a)



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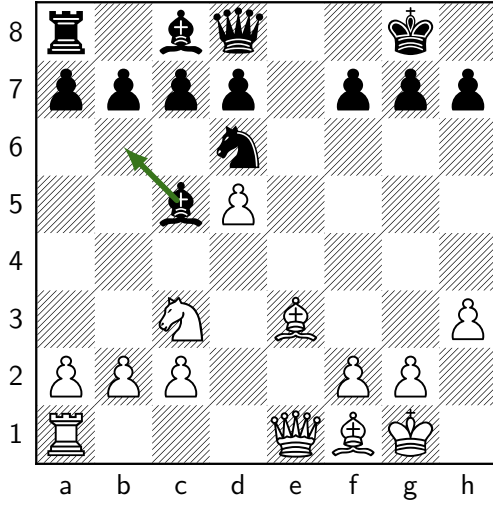


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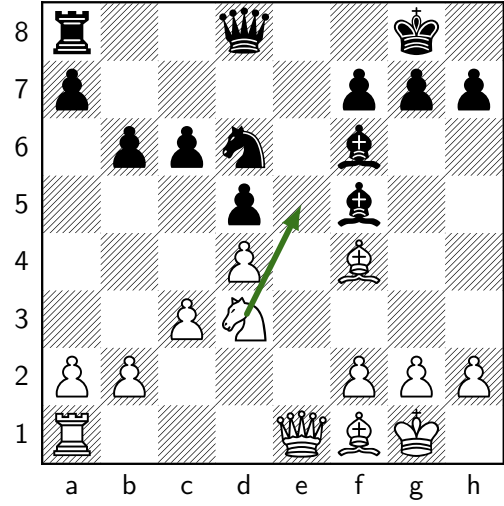


(d)

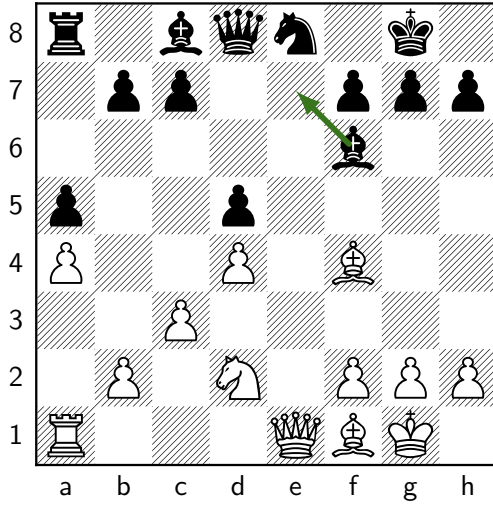
Figure 6: **Move 15, Black: Nd6.** (a) Our position; Melanie plays Nd6 (e8d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#3158` (ply 27) [structural distance $d=48$, QAP obj=43]: the master played Nd6; under μ our *Ne1* plays the role of that game’s *Ne1*; cross-type roles: our *Bc8* as their *Ra8*, our *nc3* as their *Bc8*; transferred move e8d6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#1415` (ply 27) [structural distance $d=48$, QAP obj=43]: the master played Nd6; under μ our *Ne1* plays the role of that game’s *Ne1*; cross-type roles: our *Bc8* as their *Ra8*, our *nc3* as their *Bc8*; transferred move e8d6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#2436` (ply 25) [structural distance $d=52$, QAP obj=25]: the master played Ne8; under μ our *Ne1* plays the role of that game’s *Nd3*; cross-type roles: our *pb2* as their *Ra8*, our *ph3* as their *bf4*; transferred move e8d6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nd6 (summed QAP objective 137, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



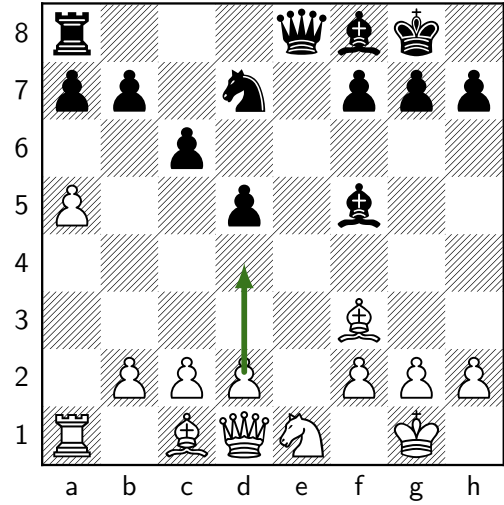
(a)



(b)

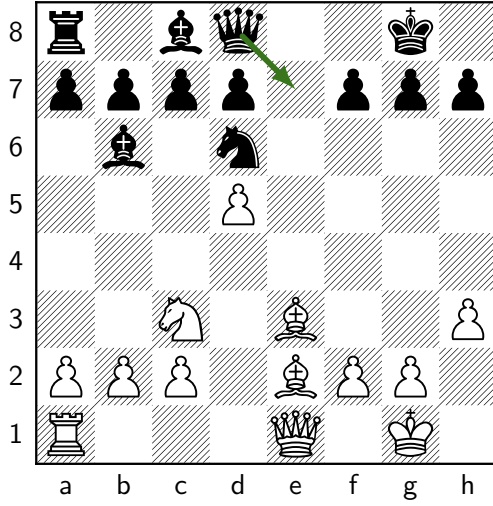


(c)

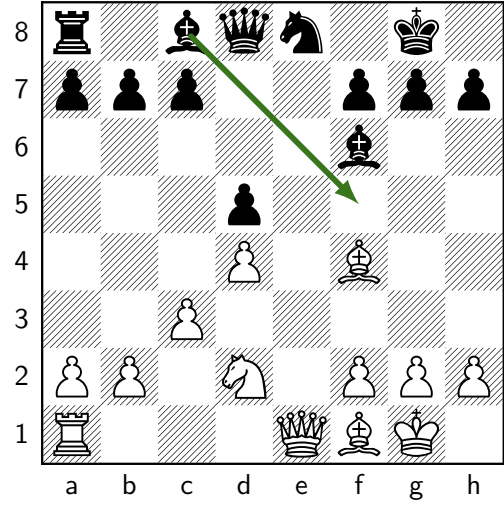


(d)

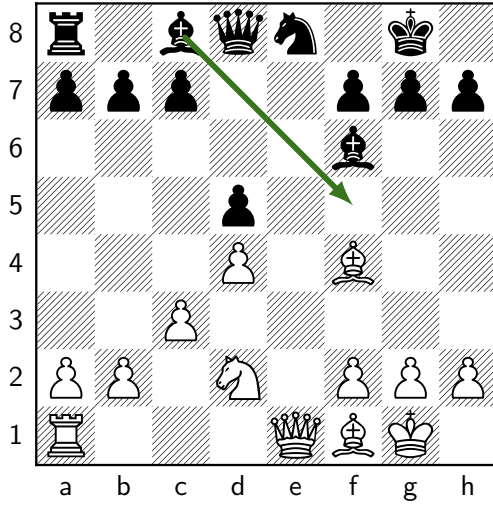
Figure 7: **Move 16, Black: Bb6**. (a) Our position; Melanie plays Bb6 (c5b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#2215` (ply 34) [structural distance $d=44$, QAP obj=34]: the master played N e5; under μ our Bc4 plays the role of that game’s Nd3 (a **cross-type** correspondence); cross-type roles: our Nd6 as their Qe1, our Pg7 as their Bf1; transferred move c5b6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#805` (ply 29) [structural distance $d=47$, QAP obj=11]: the master played Be7; under μ our Nd3 plays the role of that game’s Bf3 (a **cross-type** correspondence); cross-type roles: our nc3 as their Bc8, our Pc7 as their Qd8; transferred move d6e8. (d) Retrieved from `lichess_elite_2024-09_p0026#2448` (ply 28) [structural distance $d=48$, QAP obj=10]: the master played d4; under μ our Pc2 plays the role of that game’s Pd2; cross-type roles: our Bc8 as their Ra1, our Pa7 as their Bc1; transferred move c7c6. Synthesis: 1 retrieved analog transferred to Bb6 (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



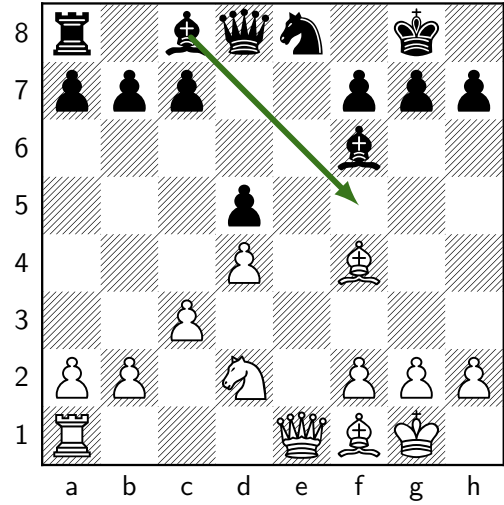
(a)



(b)

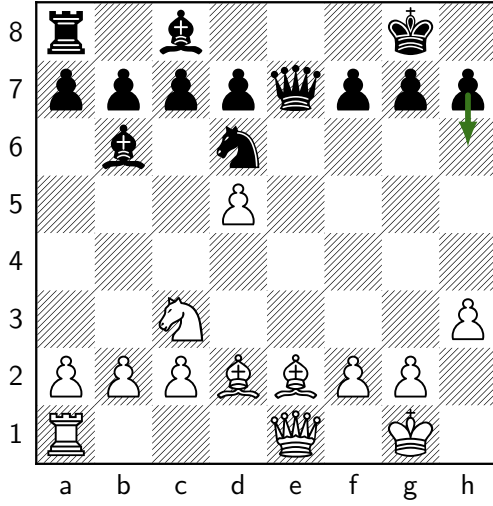


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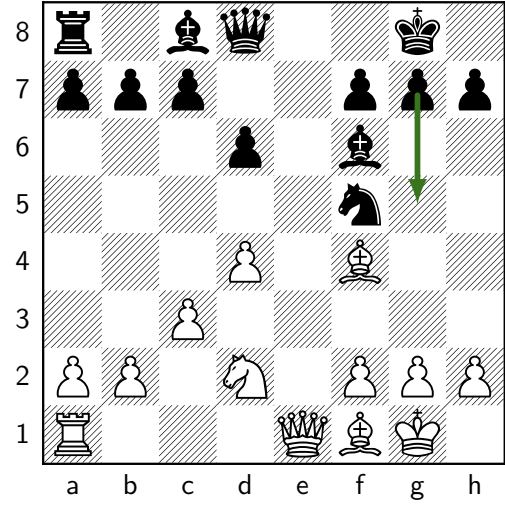


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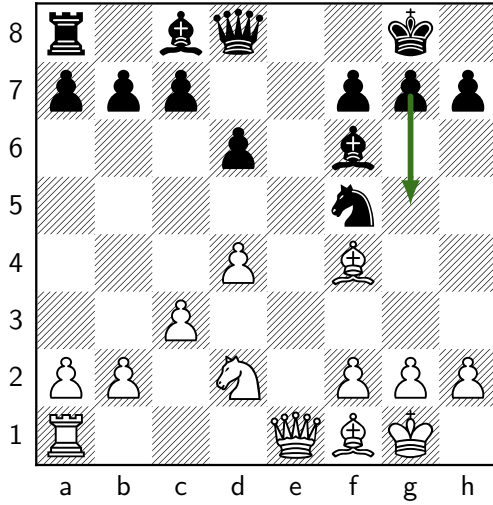
Figure 8: **Move 17, Black: Qe7.** (a) Our position; Melanie plays Qe7 (d8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0012#3688` (ply 27) [structural distance $d=71$, QAP obj=13]: the master played Bf5; under μ our Qd1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Pa7 as their Qd8, our Pc7 as their Ne8; transferred move d8e7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#327` (ply 27) [structural distance $d=71$, QAP obj=13]: the master played Bf5; under μ our Qd1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Pa7 as their Qd8, our Pc7 as their Ne8; transferred move d8e7 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0016#690` (ply 27) [structural distance $d=71$, QAP obj=13]: the master played Bf5; under μ our Qd1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Pa7 as their Qd8, our Pc7 as their Ne8; transferred move d8e7 (**endorses this move**). Synthesis: 43 retrieved analogs transferred to Qe7 (summed QAP objective 561, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



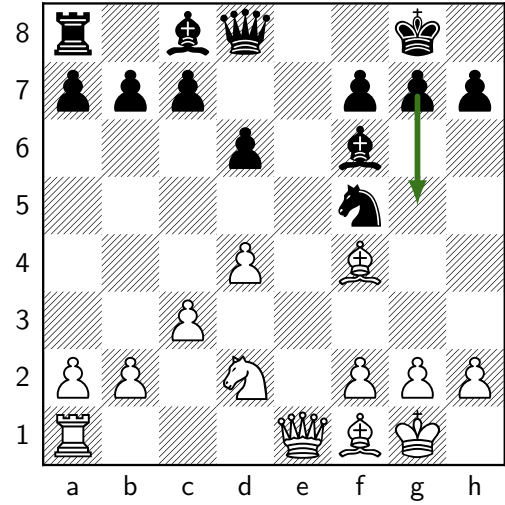
(a)



(b)

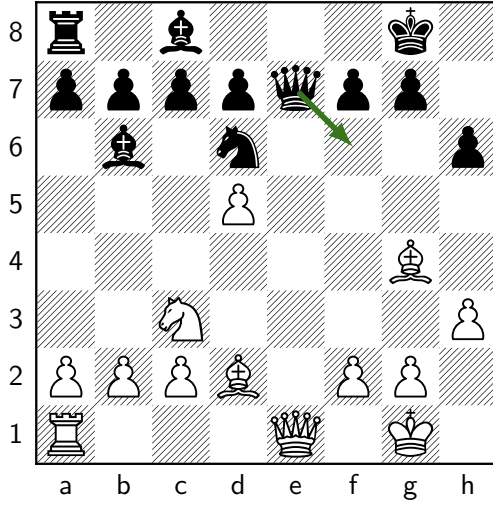


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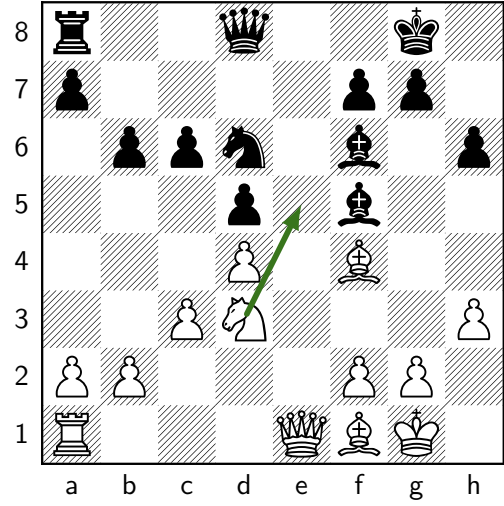


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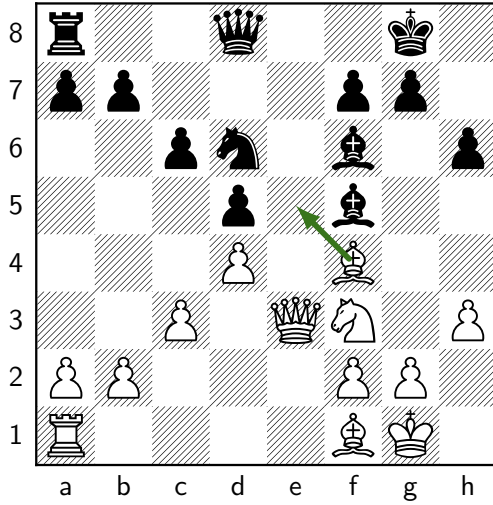
Figure 9: **Move 18, Black: h6.** (a) Our position; Melanie plays **h6** (**h7h6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#403` (ply 27) [structural distance $d=85$, QAP obj=37]: the master played **g5**; under μ our *Ph2* plays the role of that game’s *Pg2*; cross-type roles: our *pg2* as their *Ra8*, our *Pb7* as their *Bc8*; transferred move **h7h6** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#496` (ply 27) [structural distance $d=85$, QAP obj=37]: the master played **g5**; under μ our *Ph2* plays the role of that game’s *Pg2*; cross-type roles: our *pg2* as their *Ra8*, our *Pb7* as their *Bc8*; transferred move **h7h6** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#580` (ply 27) [structural distance $d=85$, QAP obj=37]: the master played **g5**; under μ our *Ph2* plays the role of that game’s *Pg2*; cross-type roles: our *pg2* as their *Ra8*, our *Pb7* as their *Bc8*; transferred move **h7h6** (**endorses this move**). Synthesis: 11 retrieved analogs transferred to **h6** (summed QAP objective 348, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



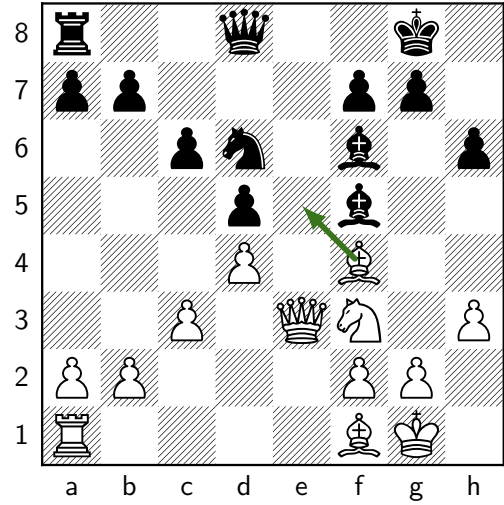
(a)



(b)

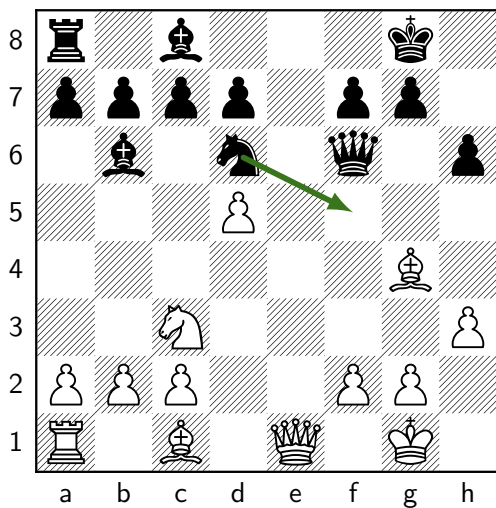


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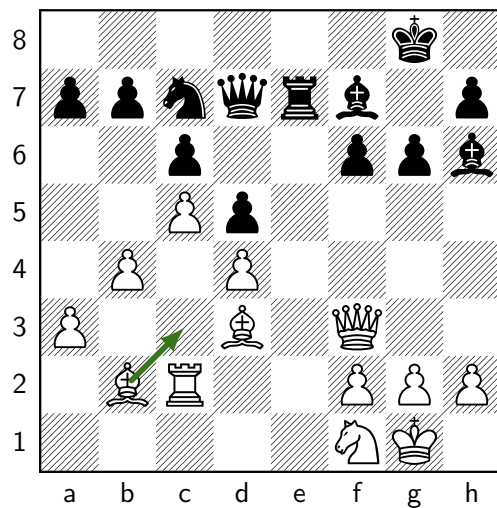


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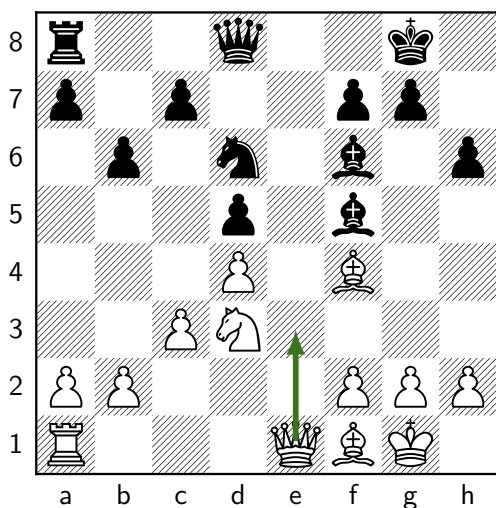
Figure 10: **Move 19, Black: Qf6 (e7f6)**. (a) Our position; Melanie plays Qf6 (e7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#1006` (ply 36) [structural distance $d=71$, QAP obj=34]: the master played Ne5; under μ our Qe2 plays the role of that game’s Nd3 (a **cross-type** correspondence); cross-type roles: our Bb6 as their Qe1, our Nd6 as their Pc3; transferred move e7f6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#3526` (ply 34) [structural distance $d=70$, QAP obj=27]: the master played Be5; under μ our Qe2 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Bf1, our Bb6 as their Pb2; transferred move e7f6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#3493` (ply 34) [structural distance $d=70$, QAP obj=27]: the master played Be5; under μ our Qe2 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Bf1, our Bb6 as their Pb2; transferred move e7f6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qf6 (summed QAP objective 135, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



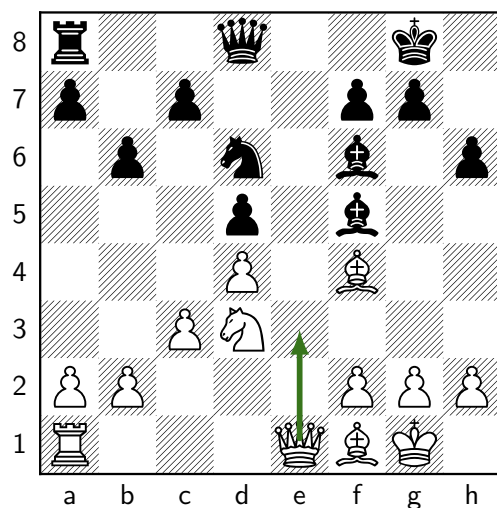
(a)



(b)

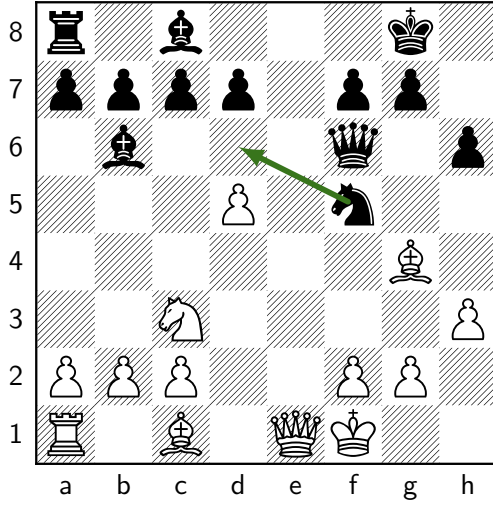


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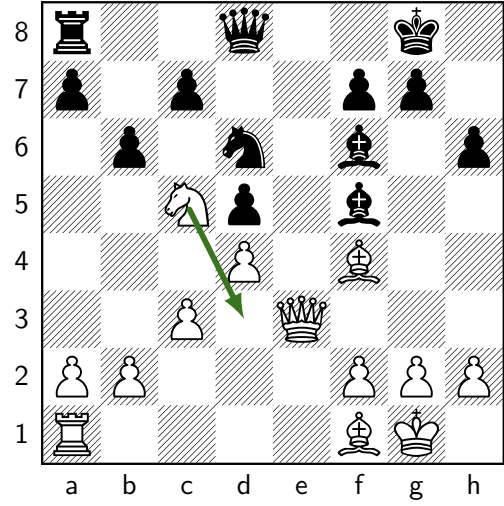


(d)

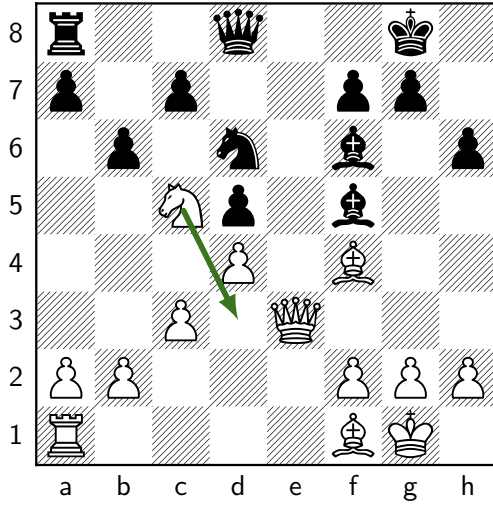
Figure 11: **Move 20, Black: Nf5.** (a) Our position; Melanie plays Nf5 (d6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#1942` (ply 44) [structural distance $d=96$, QAP obj=40]: the master played Bc3; under μ our Nd3 plays the role of that game’s Bb2 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Nf1, our pd5 as their Rc2; transferred move d6f5 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#75` (ply 34) [structural distance $d=95$, QAP obj=24]: the master played Qe3; under μ our Nd3 plays the role of that game’s Qe1 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Bf1, our Bb6 as their Pb2; transferred move d6f5 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#793` (ply 34) [structural distance $d=95$, QAP obj=24]: the master played Qe3; under μ our Nd3 plays the role of that game’s Qe1 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Bf1, our Bb6 as their Pb2; transferred move d6f5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Nf5 (summed QAP objective 182, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



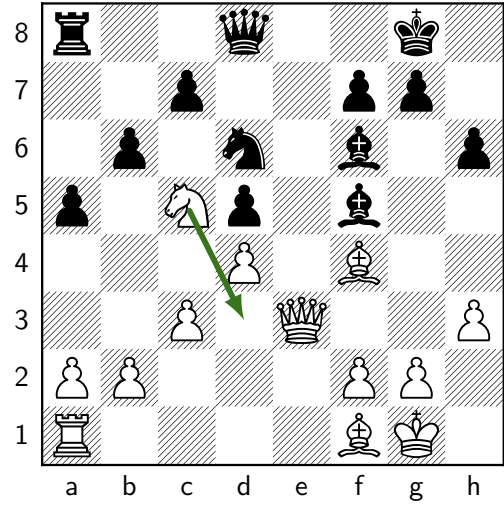
(a)



(b)

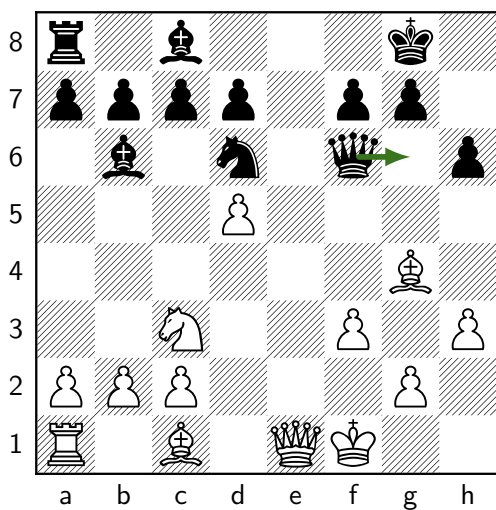


(c)

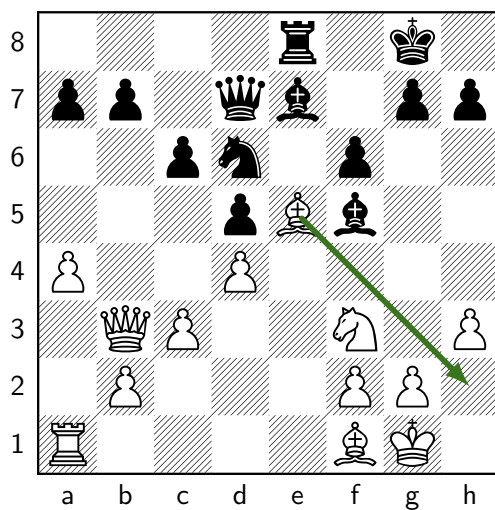


(d)

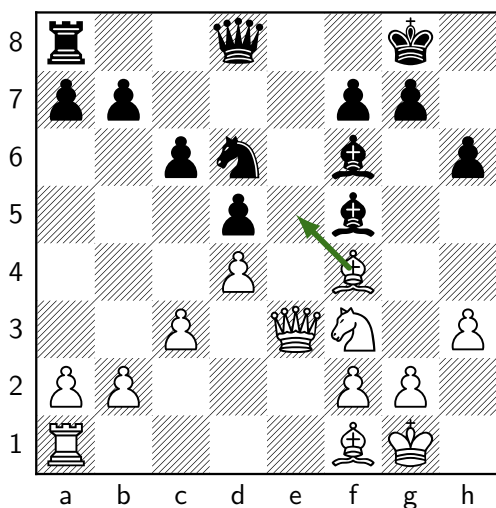
Figure 12: **Move 21, Black: Nd6.** (a) Our position; Melanie plays Nd6 (f5d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#1972` (ply 34) [structural distance $d=65$, QAP obj=29]: the master played Nd3; under μ our $Nf4$ plays the role of that game’s $Nc5$; cross-type roles: our $ra1$ as their $Bf1$, our $Bc8$ as their $Pa2$; transferred move f5d6 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#3049` (ply 34) [structural distance $d=65$, QAP obj=29]: the master played Nd3; under μ our $Nf4$ plays the role of that game’s $Nc5$; cross-type roles: our $ra1$ as their $Bf1$, our $Bc8$ as their $Pa2$; transferred move f5d6 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#3414` (ply 36) [structural distance $d=69$, QAP obj=27]: the master played Nd3; under μ our $Nf4$ plays the role of that game’s $Nc5$; cross-type roles: our $Bc8$ as their $Ra1$, our $ra1$ as their $Bf1$; transferred move f5d6 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Nd6 (summed QAP objective 232, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



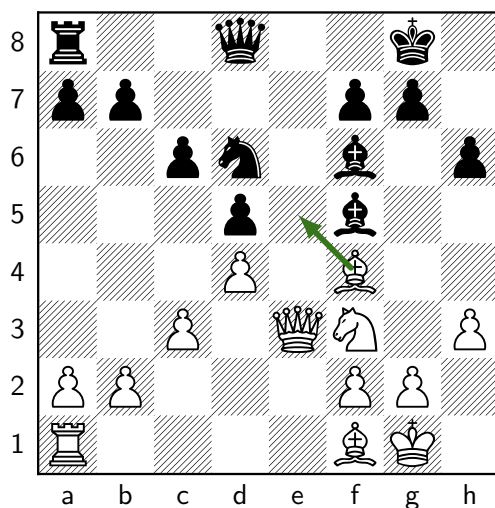
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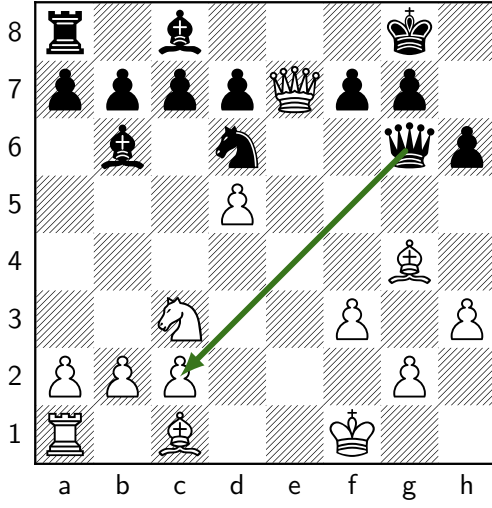


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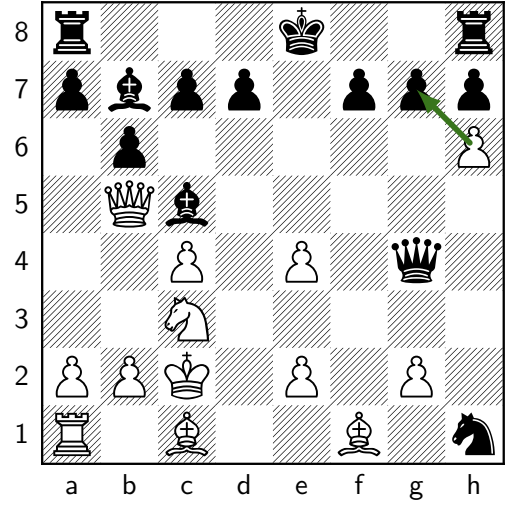


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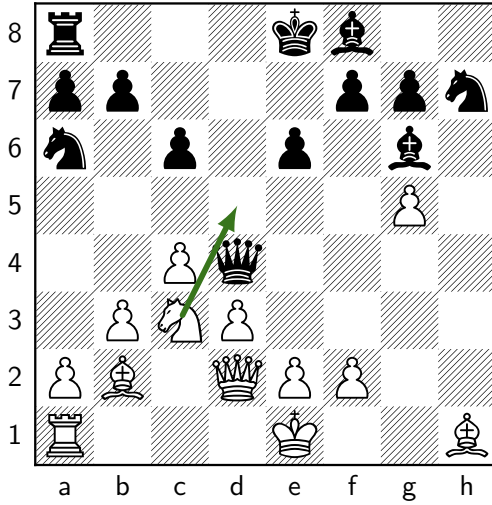
Figure 13: **Move 22, Black: Qg6**. (a) Our position; Melanie plays Qg6 (f6g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2025-11_p0000#3737 (ply 42) [structural distance $d=102$, QAP obj=31]: the master played Bh2; under μ our Qf3 plays the role of that game’s Be5 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra1, our pd5 as their Bf1; transferred move f6g6 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0005#3526 (ply 34) [structural distance $d=79$, QAP obj=28]: the master played Be5; under μ our Qf3 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our ra1 as their Bf1, our Bc8 as their Pa2; transferred move f6g6 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0019#3493 (ply 34) [structural distance $d=79$, QAP obj=28]: the master played Be5; under μ our Qf3 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our ra1 as their Bf1, our Bc8 as their Pa2; transferred move f6g6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qg6 (summed QAP objective 116, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



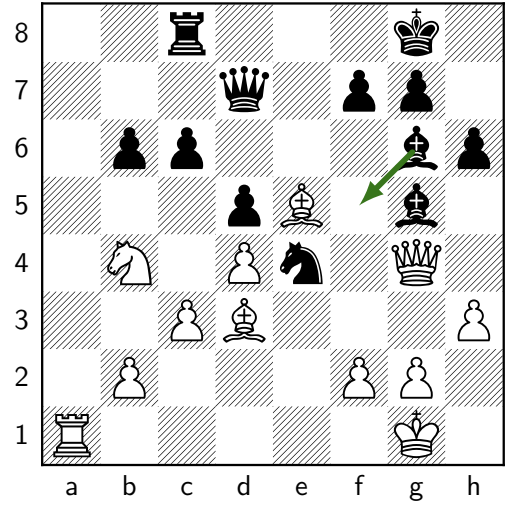
(a)



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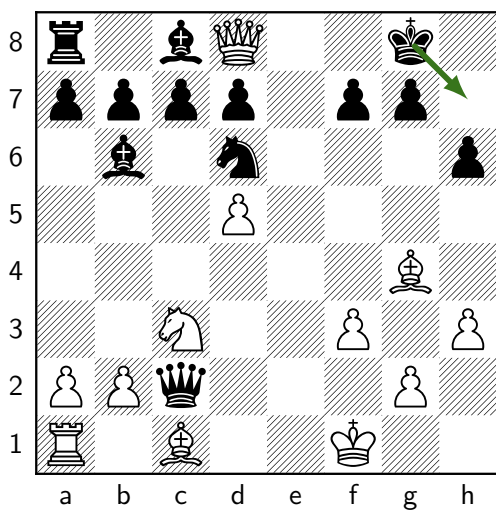


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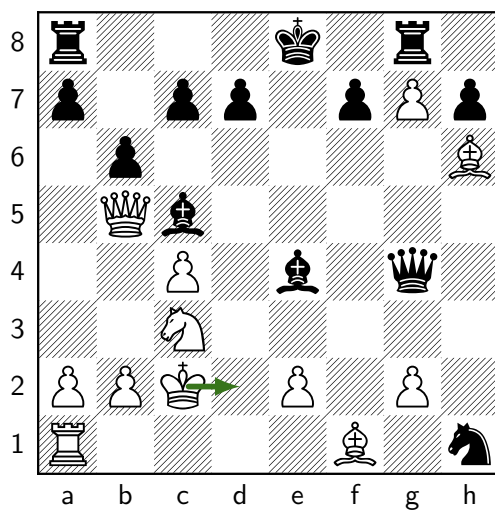


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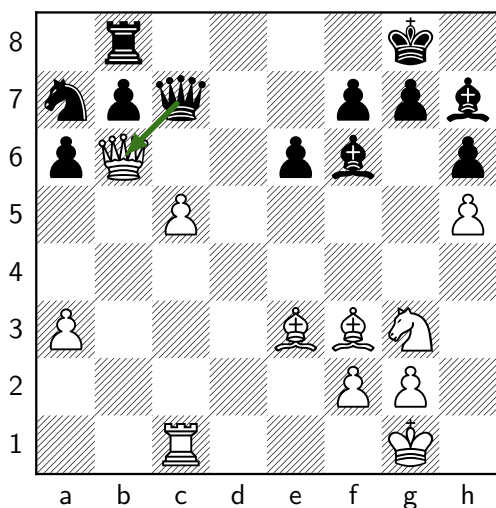
Figure 14: **Move 23, Black: Qxc2.** (a) Our position; Melanie plays Qxc2 (g6c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#1838` (ply 32) [structural distance $d=127$, QAP obj=50]: the master played hxg7; under μ our Qg3 plays the role of that game’s Ph6 (a **cross-type** correspondence); cross-type roles: our nc3 as their Bc1, our Ra8 as their Bf1; transferred move g6c2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#165` (ply 28) [structural distance $d=120$, QAP obj=59]: the master played Nd5; under μ our Nd3 plays the role of that game’s Nc3; cross-type roles: our Ph6 as their Bh1, our Pg7 as their Bb2; transferred move d6e4. (d) Retrieved from `lichess_elite_2025-11_p0000#2045` (ply 57) [structural distance $d=129$, QAP obj=58]: the master played Bf5; under μ our Qg3 plays the role of that game’s Bg3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Rc8, our Nd6 as their Qd7; transferred move g6f5. Synthesis: 1 retrieved analog transferred to Qxc2 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



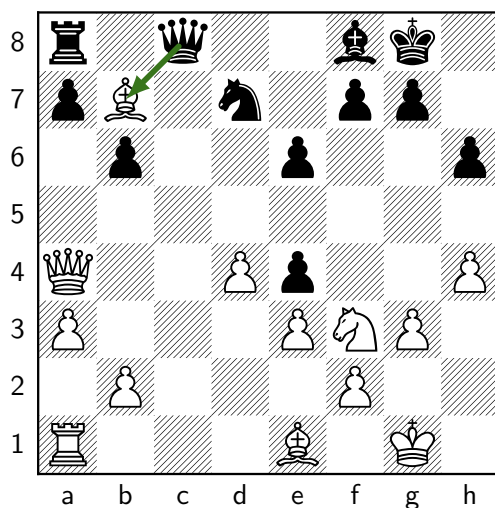
(a)



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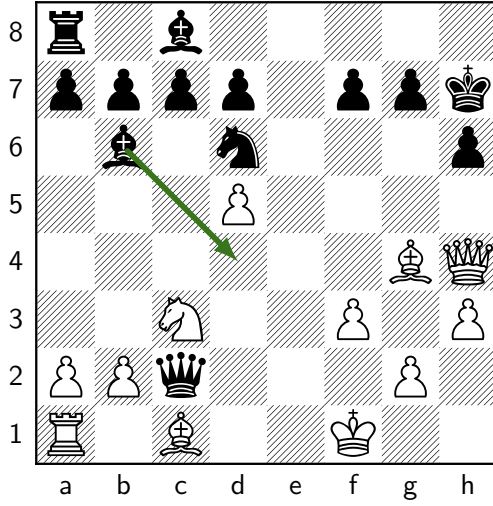


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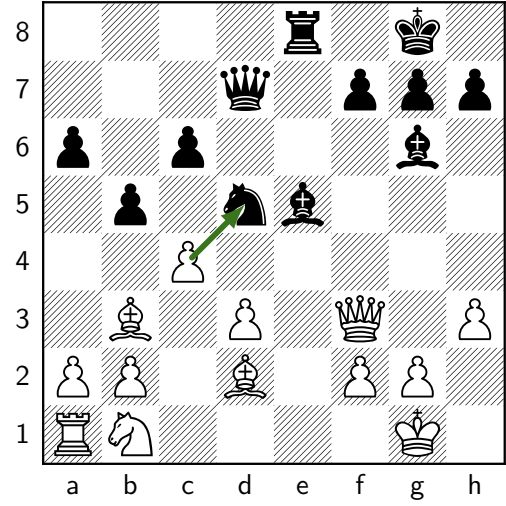


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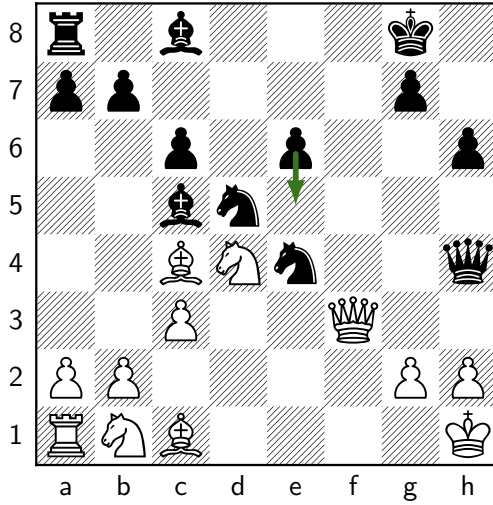
Figure 15: **Move 24, Black: Kh7.** (a) Our position; Melanie plays Kh7 (g8h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0019#1838 (ply 36) [structural distance $d=124$, QAP obj=51]: the master played Kd2; under μ our Kg1 plays the role of that game’s Kc2; cross-type roles: our Pd7 as their nh1, our Bb6 as their Pa2; transferred move g8h7 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0010#567 (ply 59) [structural distance $d=146$, QAP obj=26]: the master played Qxb6; under μ our Kg1 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Nd6 as their Rb8, our Bc8 as their Na7; transferred move g8h7 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0016#3684 (ply 45) [structural distance $d=172$, QAP obj=8]: the master played Qxb7; under μ our Kg1 plays the role of that game’s Qc1 (a **cross-type** correspondence); cross-type roles: our Bb6 as their Ra8, our Pf7 as their Bf8; transferred move g8h7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kh7 (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



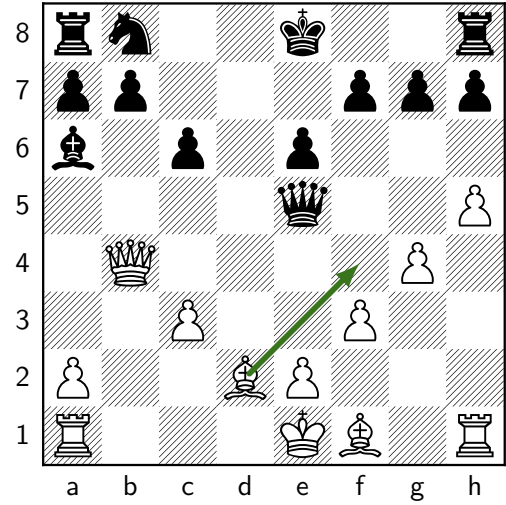
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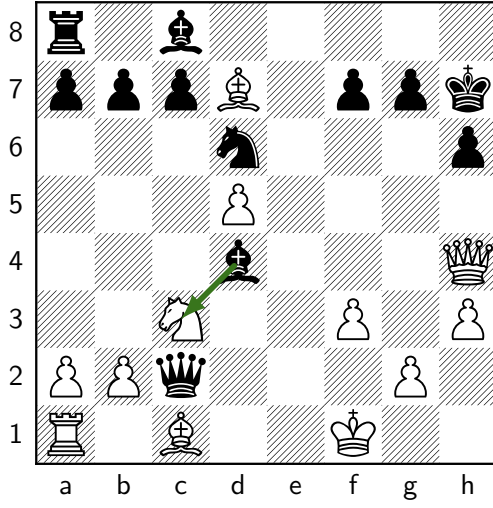


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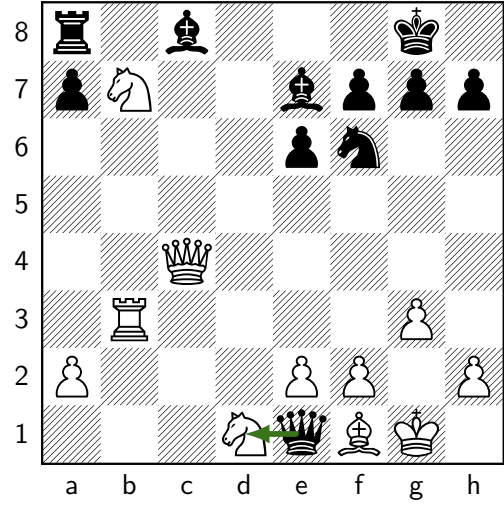


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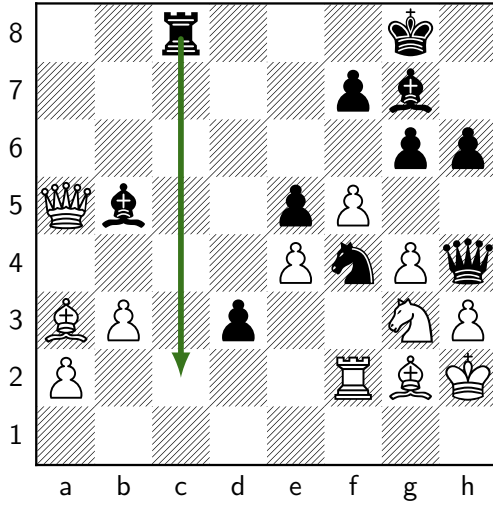
Figure 16: **Move 25, Black: Bd4**. (a) Our position; Melanie plays Bd4 (b6d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#1543` (ply 36) [structural distance $d=108$, QAP obj=70]: the master played `cx d5`; under μ our *Qc7* plays the role of that game’s *Pc4* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra1*, our *Pf7* as their *Nb1*; transferred move `c2e4`. (c) Retrieved from `lichess_elite_2024-09_p0019#137` (ply 33) [structural distance $d=130$, QAP obj=66]: the master played `e5`; under μ our *Nd3* plays the role of that game’s *Pe3* (a **cross-type** correspondence); cross-type roles: our *ph3* as their *Bc8*, our *Bc8* as their *Pb7*; transferred move `d6c4`. (d) Retrieved from `lichess_elite_2024-09_p0013#3595` (ply 32) [structural distance $d=129$, QAP obj=60]: the master played `Bf4`; under μ our *Bb3* plays the role of that game’s *Bd2*; cross-type roles: our *pd5* as their *Ra1*, our *Pg7* as their *Bf1*; transferred move `b6c5`. Synthesis: no retrieved analog transferred to Bd4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



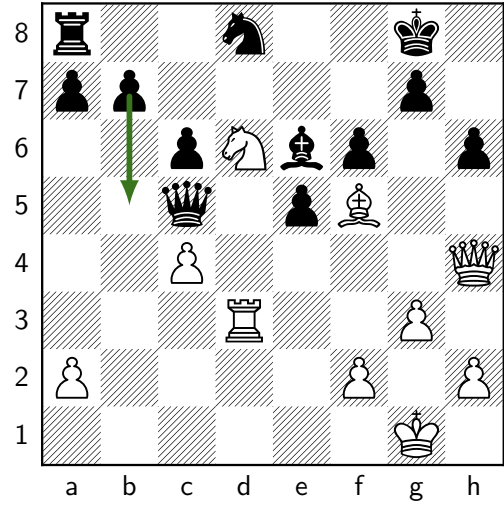
(a)



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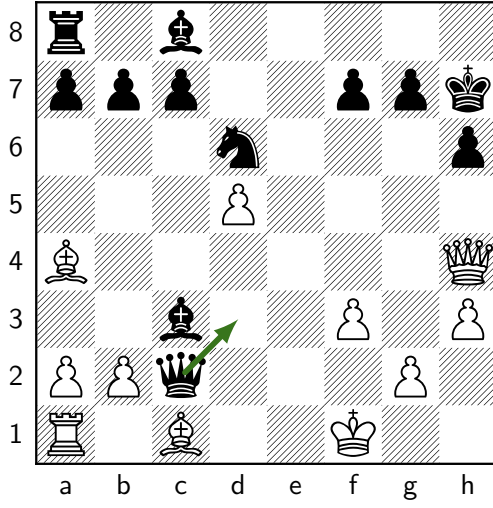


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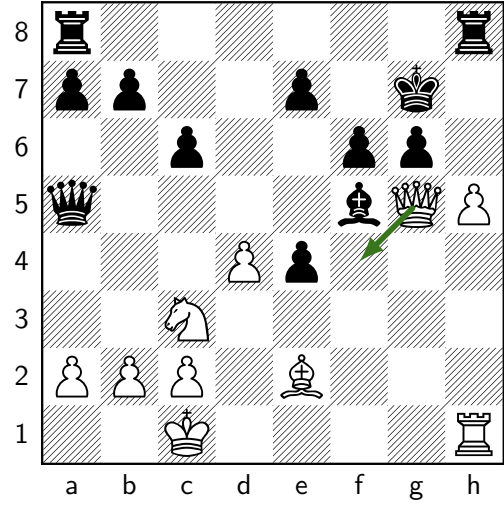


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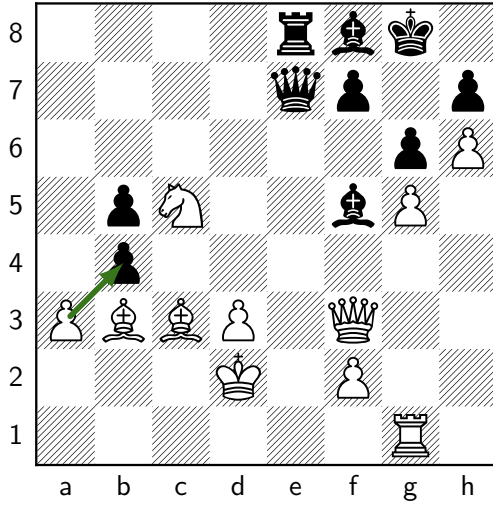
Figure 17: **Move 26, Black: Bxc3.** (a) Our position; Melanie plays Bxc3 (d4c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#2599` (ply 43) [structural distance $d=122$, QAP obj=115]: the master played Qxd1; under μ our Qc7 plays the role of that game’s Qe8; cross-type roles: our Nd6 as their Ra8, our bd7 as their nb7; transferred move c2c1. (c) Retrieved from `lichess_elite_2025-11_p0000#3372` (ply 53) [structural distance $d=117$, QAP obj=105]: the master played Rc2; under μ our Nd3 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Bg7, our bd7 as their Pg6; transferred move d6b5. (d) Retrieved from `lichess_elite_2024-09_p0020#1671` (ply 61) [structural distance $d=89$, QAP obj=103]: the master played b5; under μ our Pf2 plays the role of that game’s Pb2; cross-type roles: our Pa7 as their Ra8, our bc1 as their nd6; transferred move f7f5. Synthesis: no retrieved analog transferred to Bxc3 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



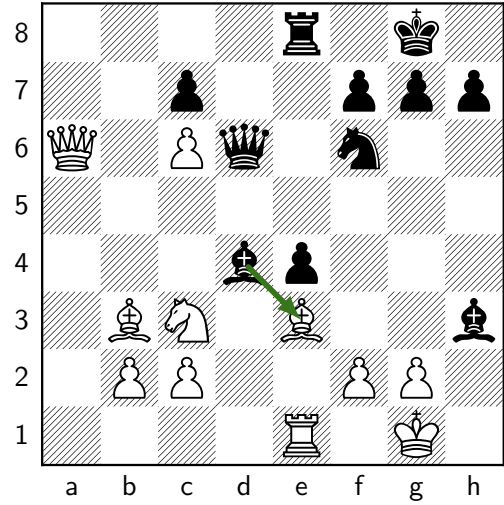
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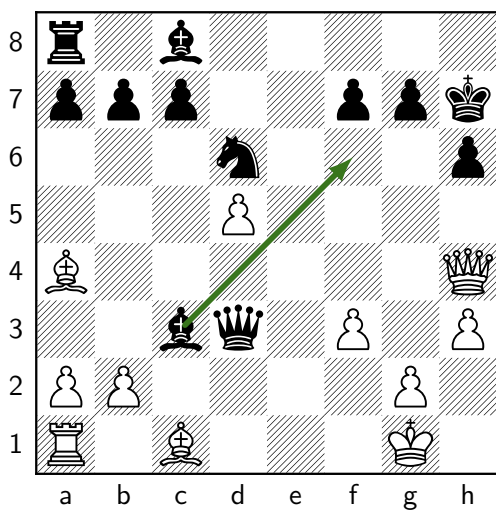


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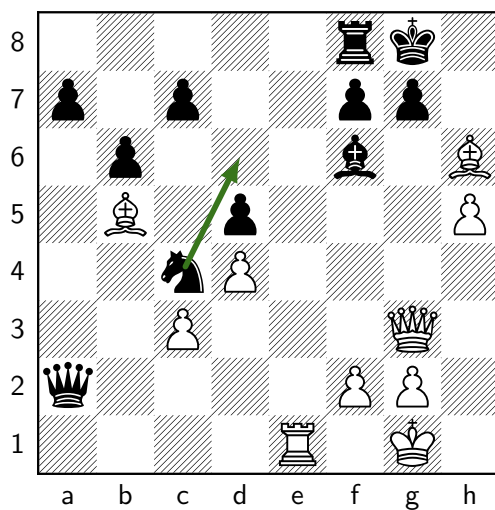


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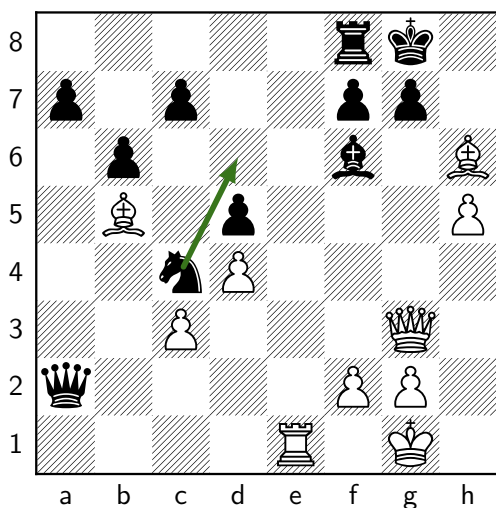
Figure 18: **Move 27, Black: Qd3+.** (a) Our position; Melanie plays Qd3+ (c2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0012#219` (ply 38) [structural distance $d=103$, QAP obj=38]: the master played Qf4; under μ our Qc7 plays the role of that game’s Qg5; cross-type roles: our Pa7 as their Rh1, our Pf7 as their Nc3; transferred move c2d3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#1654` (ply 62) [structural distance $d=107$, QAP obj=36]: the master played axb4; under μ our Qc7 plays the role of that game’s Pa3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Rg1, our qh4 as their Bb3; transferred move c2d3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#2493` (ply 39) [structural distance $d=93$, QAP obj=69]: the master played Bxe3; under μ our Qc7 plays the role of that game’s Bd5 (a **cross-type** correspondence); cross-type roles: our pd5 as their Re8, our Nd6 as their qa6; transferred move c2e4. Synthesis: 3 retrieved analogs transferred to Qd3+ (summed QAP objective 93, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



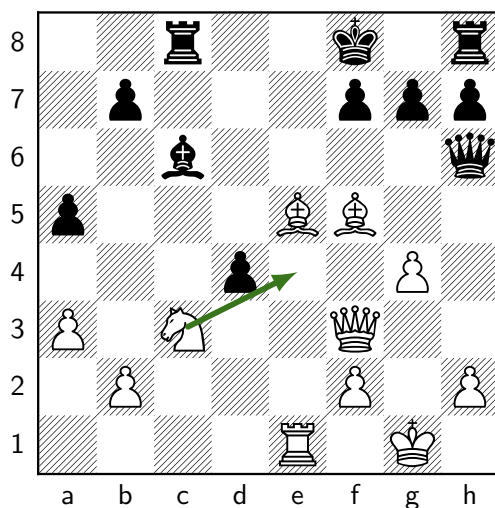
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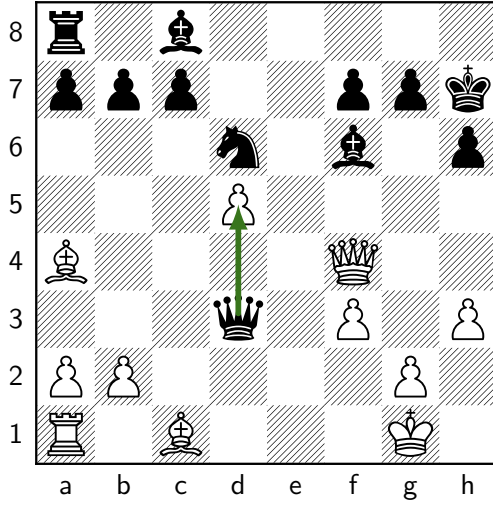


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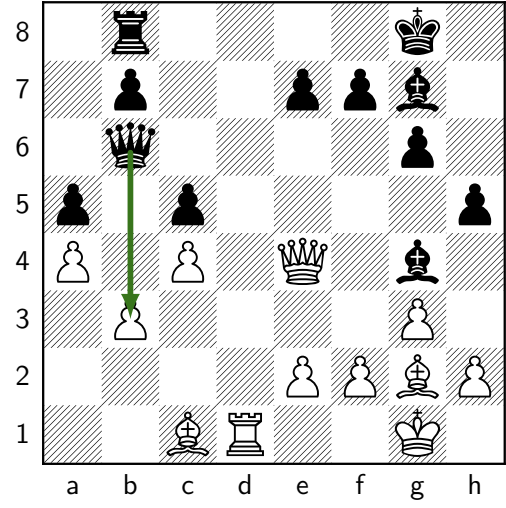


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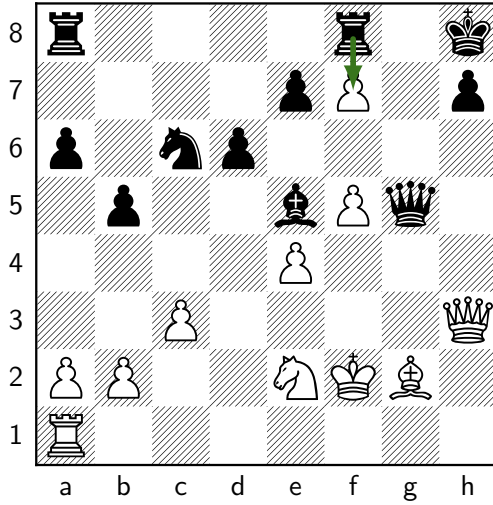
Figure 19: **Move 28, Black: Bf6**. (a) Our position; Melanie plays Bf6 (c3f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#793` (ply 55) [structural distance $d=83$, QAP obj=57]: the master played Nd6; under μ our Bc6 plays the role of that game’s Nc5 (a **cross-type** correspondence); cross-type roles: our Qd3 as their Rf8, our qh4 as their Bf6; transferred move c3f6 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#2922` (ply 55) [structural distance $d=83$, QAP obj=57]: the master played Nd6; under μ our Bc6 plays the role of that game’s Nc5 (a **cross-type** correspondence); cross-type roles: our Qd3 as their Rf8, our qh4 as their Bf6; transferred move c3f6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#2877` (ply 54) [structural distance $d=86$, QAP obj=47]: the master played Ne4; under μ our Bc6 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Qd3 as their Re1, our Ra8 as their Pf2; transferred move c3f6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bf6 (summed QAP objective 201, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



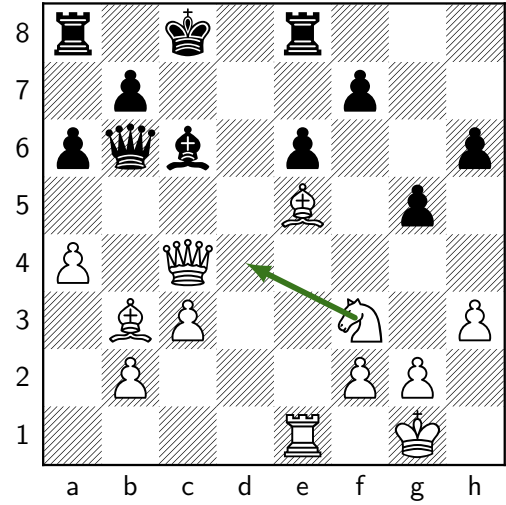
(a)



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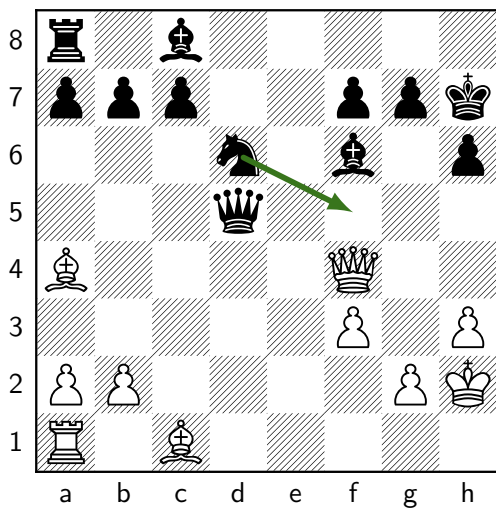


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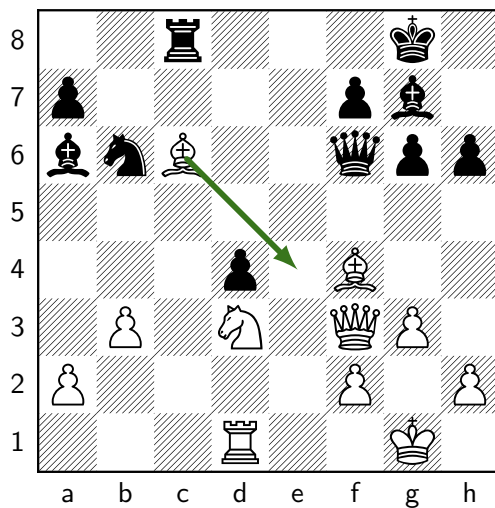


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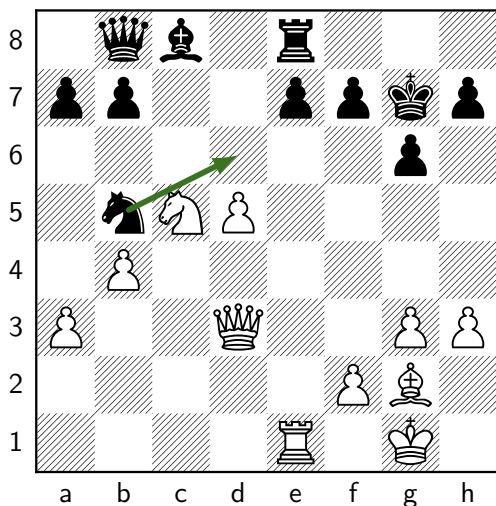
Figure 20: **Move 29, Black: Qxd5.** (a) Our position; Melanie plays Qxd5 (d3d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#2414` (ply 47) [structural distance $d=107$, QAP obj=70]: the master played Qxb3; under μ our Qd6 plays the role of that game’s Qb3; cross-type roles: our Nd6 as their Pe7, our bc1 as their pc4; transferred move d3e2. (c) Retrieved from `lichess_elite_2024-09_p0025#533` (ply 55) [structural distance $d=109$, QAP obj=57]: the master played Rxf7; under μ our Qd6 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra8, our bc1 as their Pe7; transferred move d3a6. (d) Retrieved from `lichess_elite_2024-09_p0026#2562` (ply 50) [structural distance $d=107$, QAP obj=57]: the master played Nd4; under μ our Qd6 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our bc1 as their Pb2, our pf3 as their Bb3; transferred move d3f5. Synthesis: no retrieved analog transferred to Qxd5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



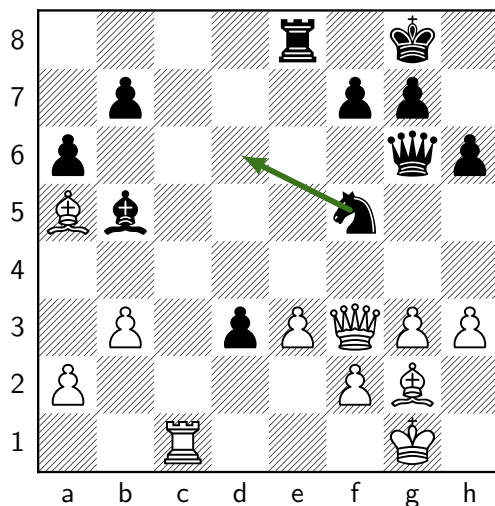
(a)



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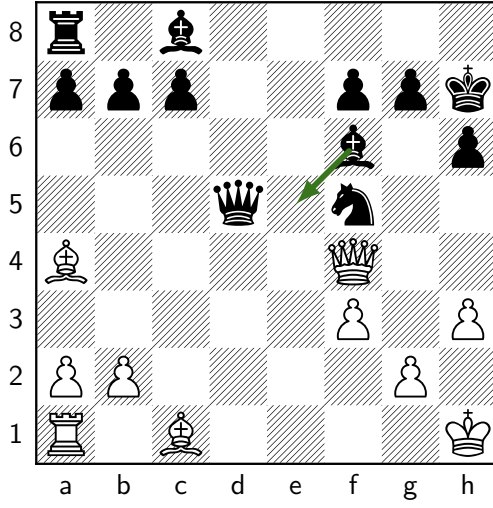


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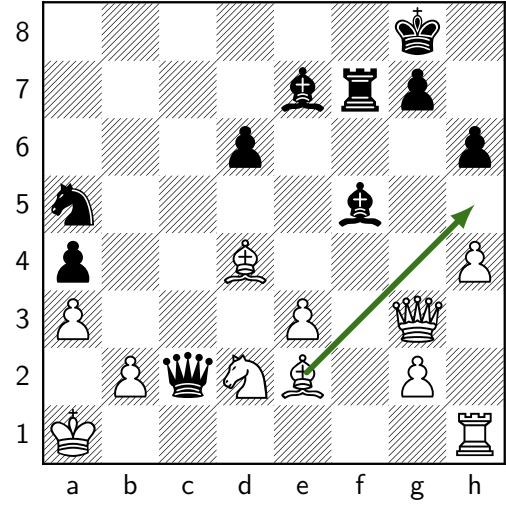


(d)

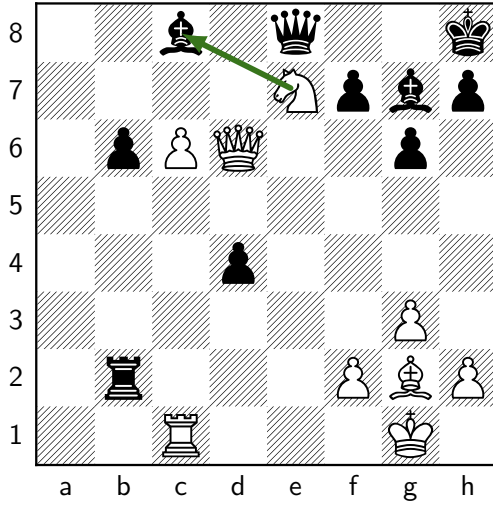
Figure 21: **Move 30, Black: Nf5**. (a) Our position; Melanie plays Nf5 (d6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#759` (ply 52) [structural distance $d=102$, QAP obj=72]: the master played Be4; under μ our Nd3 plays the role of that game’s Bc6 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Rd1, our Ra8 as their Pa2; transferred move d6f5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#3970` (ply 55) [structural distance $d=98$, QAP obj=57]: the master played Nd6; under μ our Nd3 plays the role of that game’s Nb4; cross-type roles: our ra1 as their Qb8, our pa2 as their Bc8; transferred move d6f5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#3580` (ply 63) [structural distance $d=88$, QAP obj=45]: the master played Nd6; under μ our Nd3 plays the role of that game’s Nf4; cross-type roles: our Qd5 as their Re8, our bc1 as their Pb7; transferred move d6f5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nf5 (summed QAP objective 213, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



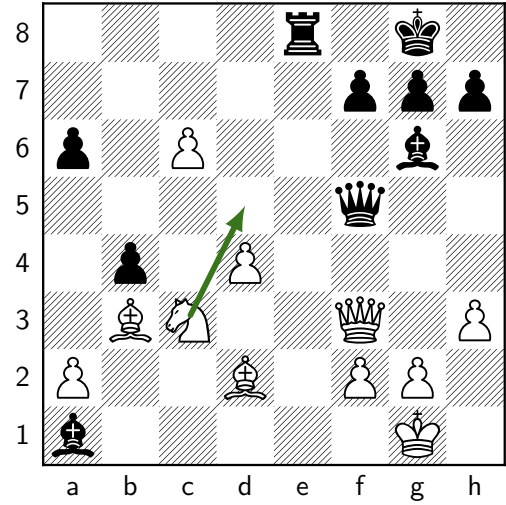
(a)



(b)

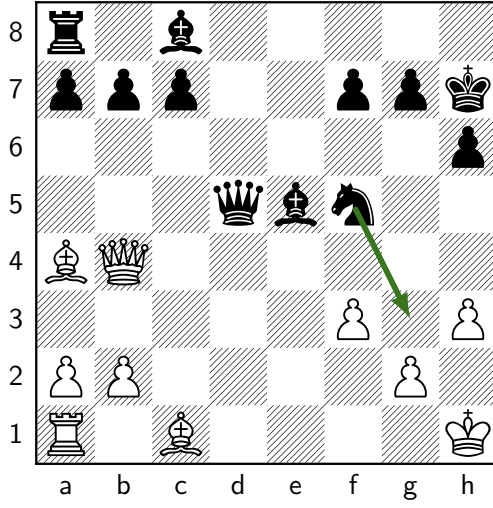


(c)

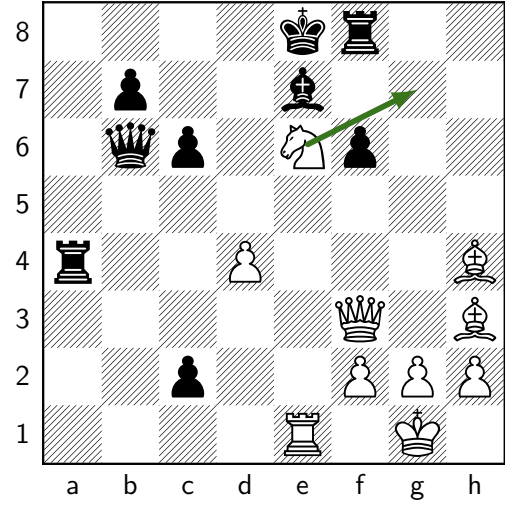


(d)

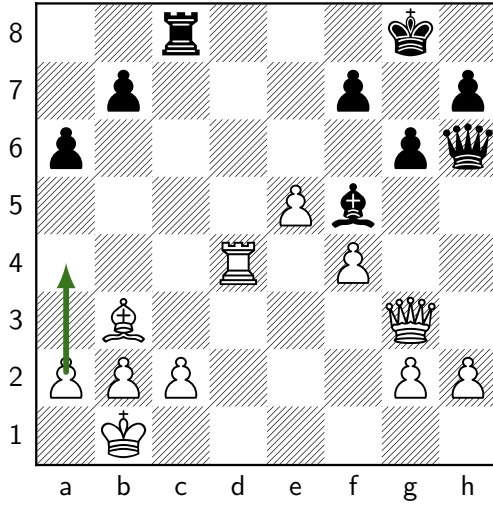
Figure 22: **Move 31, Black: Be5**. (a) Our position; Melanie plays Be5 (f6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#849` (ply 66) [structural distance $d=73$, QAP obj=53]: the master played Bh5; under μ our *Pc2* plays the role of that game’s *Be2* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Rh1*, our *Ra8* as their *Pe3*; transferred move c7c5. (c) Retrieved from `lichess_elite_2024-09_p0016#877` (ply 62) [structural distance $d=93$, QAP obj=47]: the master played Nxc8; under μ our *Nf4* plays the role of that game’s *Ne7*; cross-type roles: our *Bf6* as their *rb2*, our *Pa7* as their *Bg2*; transferred move f5g3. (d) Retrieved from `lichess_elite_2025-11_p0000#1543` (ply 44) [structural distance $d=75$, QAP obj=47]: the master played Nd5; under μ our *Nf4* plays the role of that game’s *Nc3*; cross-type roles: our *qf4* as their *ba1*, our *Qd5* as their *Bd2*; transferred move f5d4. Synthesis: no retrieved analog transferred to Be5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



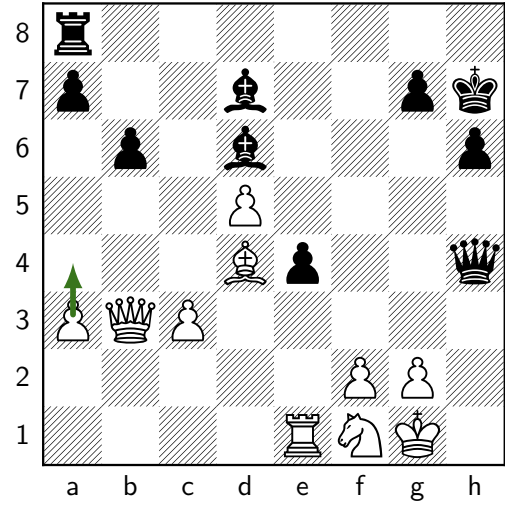
(a)



(b)

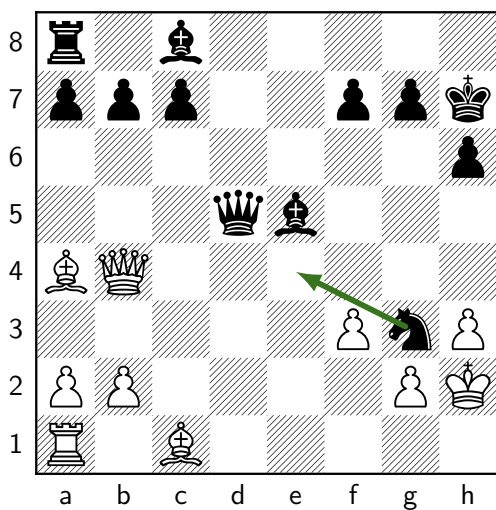


(c)

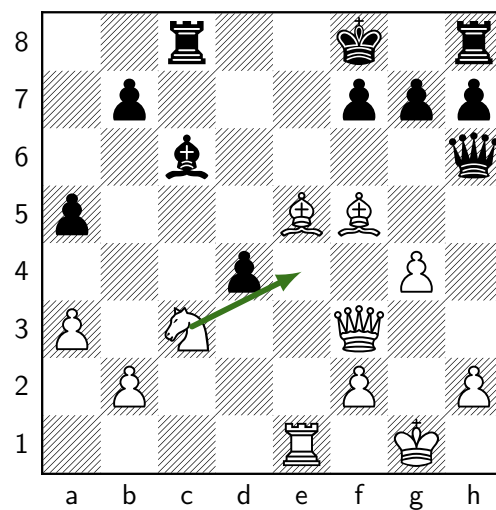


(d)

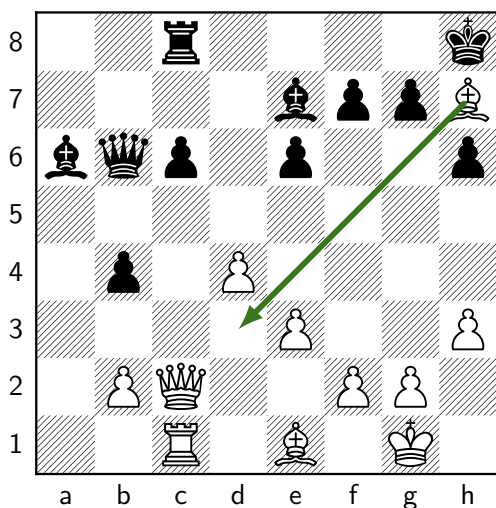
Figure 23: **Move 32, Black: Ng3+.** (a) Our position; Melanie plays Ng3+ (f5g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0012#121` (ply 62) [structural distance $d=84$, QAP obj=31]: the master played Ng7+; under μ our *Nf4* plays the role of that game’s *Ne6*; cross-type roles: our *ph3* as their *Re1*, our *Bc8* as their *pc2*; transferred move f5g3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#3717` (ply 52) [structural distance $d=106$, QAP obj=96]: the master played a4; under μ our *Pg2* plays the role of that game’s *Pa2*; cross-type roles: our *bc1* as their *Pb2*, our *ba4* as their *Pg2*; transferred move g7g5. (d) Retrieved from `lichess_elite_2024-09_p0022#3268` (ply 66) [structural distance $d=105$, QAP obj=90]: the master played a4; under μ our *Pb2* plays the role of that game’s *Pa3*; cross-type roles: our *bc1* as their *Re1*, our *Pg7* as their *Nf1*; transferred move b7b6. Synthesis: 1 retrieved analog transferred to Ng3+ (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



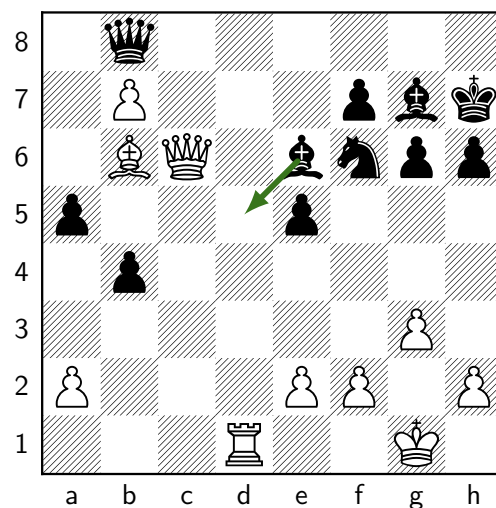
(a)



(b)

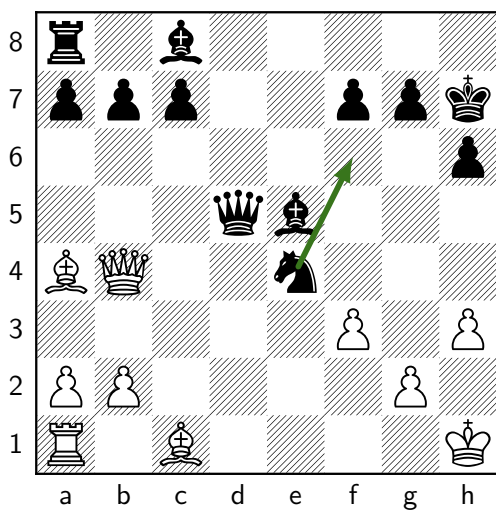


(c)

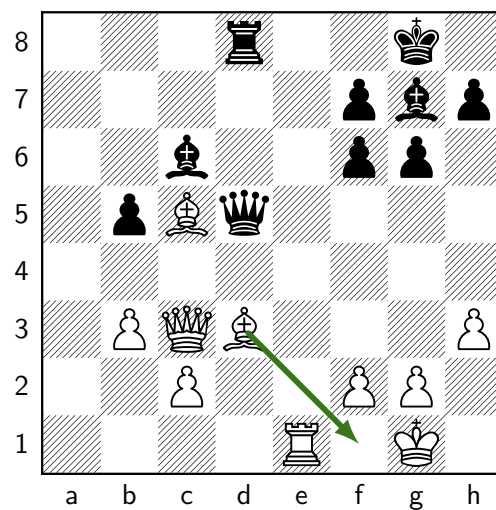


(d)

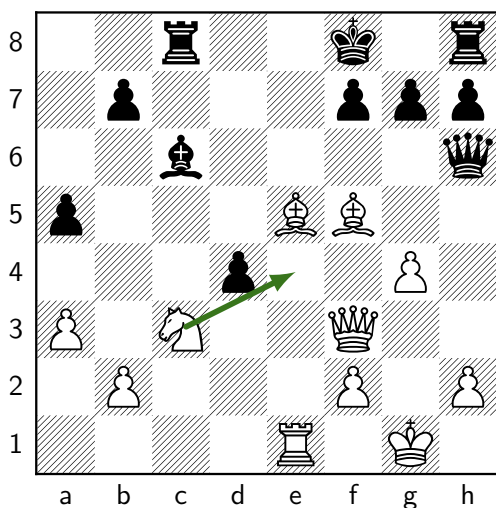
Figure 24: **Move 33, Black: Ne4+.** (a) Our position; Melanie plays Ne4+ (g3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#2877` (ply 54) [structural distance $d=81$, QAP obj=42]: the master played Ne4; under μ our Ng6 plays the role of that game’s Nc3; cross-type roles: our Bc8 as their Pa3, our Ng3 as their Qf3; transferred move g3e4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#1425` (ply 58) [structural distance $d=87$, QAP obj=42]: the master played Bd3; under μ our Ng6 plays the role of that game’s Bh7 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Rc1, our pa2 as their Be1; transferred move g3e4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#3582` (ply 51) [structural distance $d=96$, QAP obj=14]: the master played Bd5; under μ our Ng6 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our Be5 as their Qb8, our Qd5 as their Pf7; transferred move g3e4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ne4+ (summed QAP objective 99, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



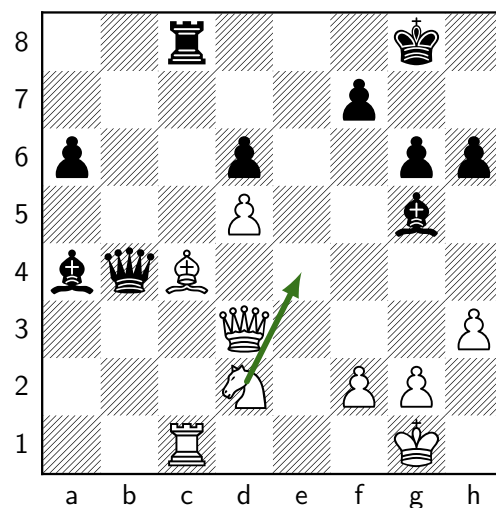
(a)



(b)

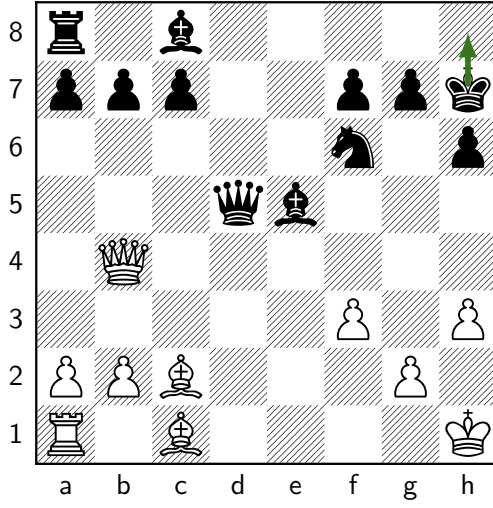


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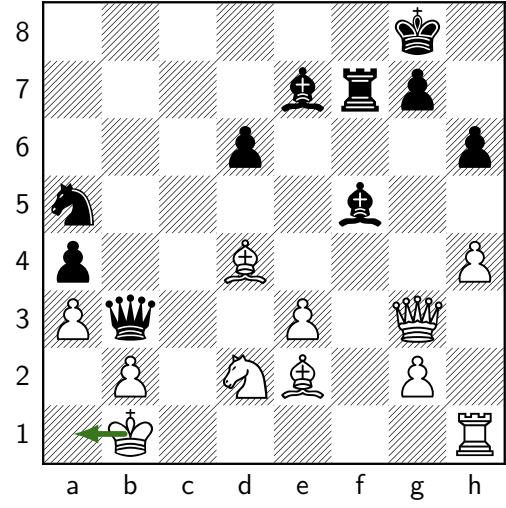


(d)

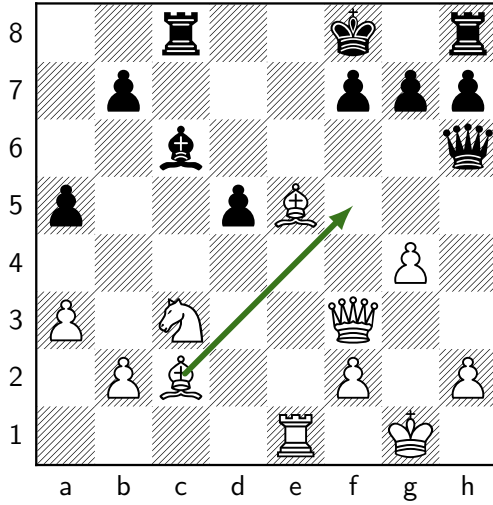
Figure 25: **Move 34, Black: Nf6**. (a) Our position; Melanie plays Nf6 (e4f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0021#3116 (ply 56) [structural distance $d=92$, QAP obj=55]: the master played Bf1; under μ our Ne5 plays the role of that game’s Bd3 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Re1, our Be5 as their Qc3; transferred move e4f6 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0022#2877 (ply 54) [structural distance $d=72$, QAP obj=46]: the master played Ne4; under μ our Ne5 plays the role of that game’s Nc3; cross-type roles: our Pg7 as their Re1, our Qd5 as their Pf2; transferred move e4f6 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0021#2826 (ply 60) [structural distance $d=80$, QAP obj=38]: the master played Ne4; under μ our Ne5 plays the role of that game’s Nd2; cross-type roles: our Be5 as their Rc1, our pb2 as their ba4; transferred move e4f6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nf6 (summed QAP objective 140, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



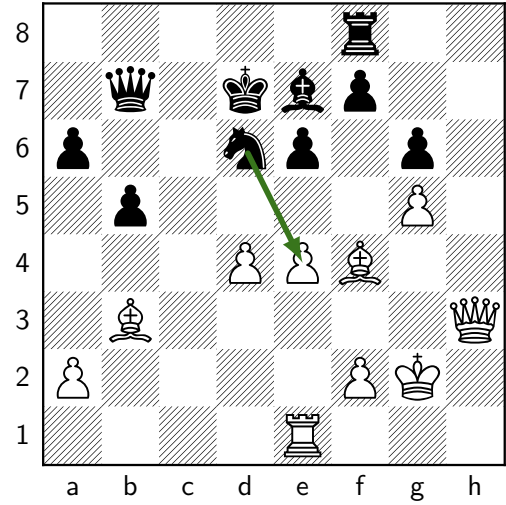
(a)



(b)

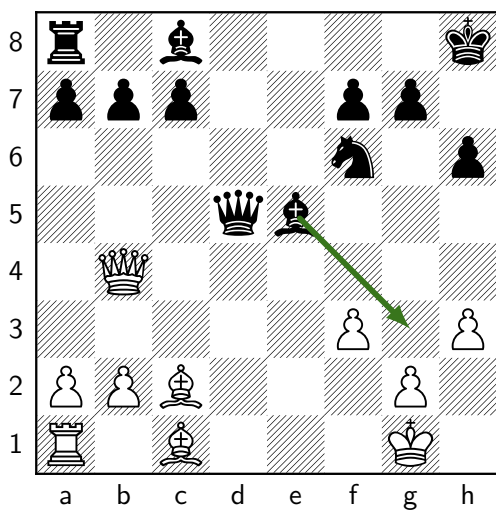


(c)

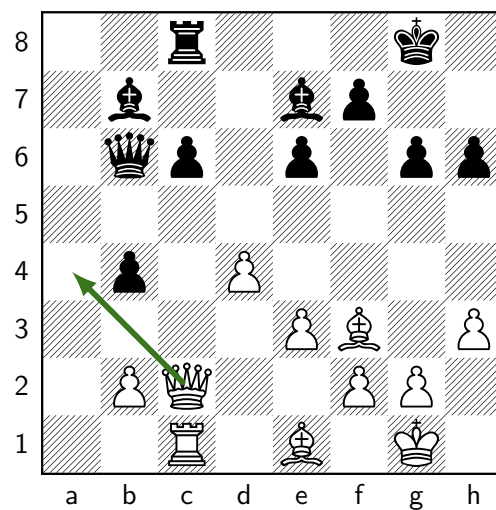


(d)

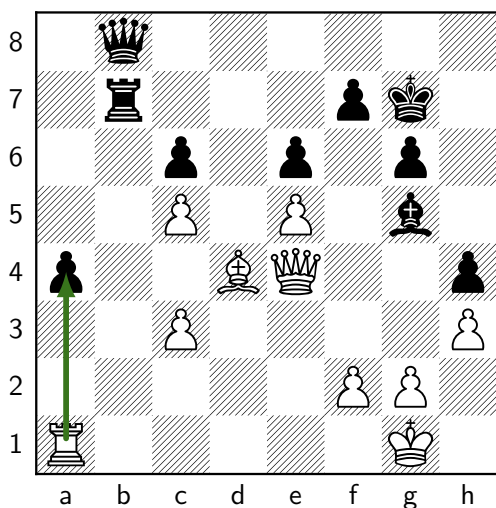
Figure 26: **Move 35, Black: Kh8**. (a) Our position; Melanie plays Kh8 (h7h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#849` (ply 64) [structural distance $d=88$, QAP obj=69]: the master played Ka1; under μ our Kh2 plays the role of that game’s Kb1; cross-type roles: our Pf7 as their Rh1, our qb4 as their Be2; transferred move h7h8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#2877` (ply 52) [structural distance $d=87$, QAP obj=37]: the master played Bf5; under μ our Bc1 plays the role of that game’s Bc2; cross-type roles: our Qd5 as their Re1, our Be5 as their Qf3; transferred move c8f5. (d) Retrieved from `lichess_elite_2024-09_p0030#1284` (ply 59) [structural distance $d=82$, QAP obj=37]: the master played Nx e4; under μ our Qd4 plays the role of that game’s Nd3 (a **cross-type** correspondence); cross-type roles: our bc1 as their Rf8, our Be5 as their Qb7; transferred move d5d3. Synthesis: 1 retrieved analog transferred to Kh8 (summed QAP objective 69, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



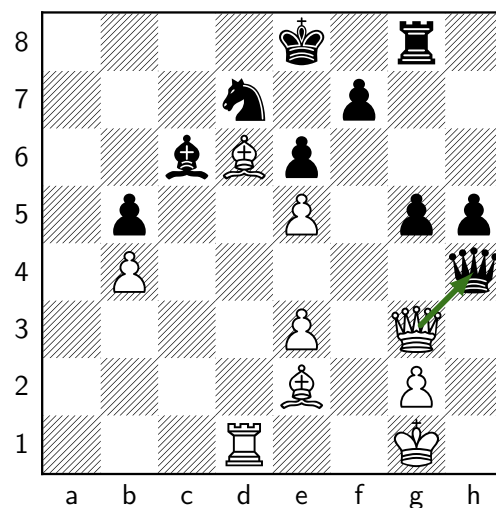
(a)



(b)

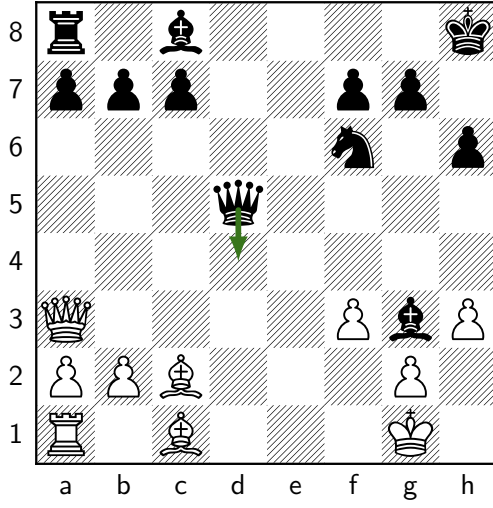


(c)

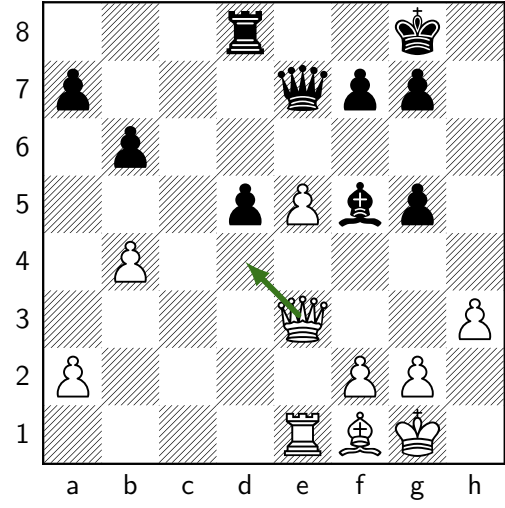


(d)

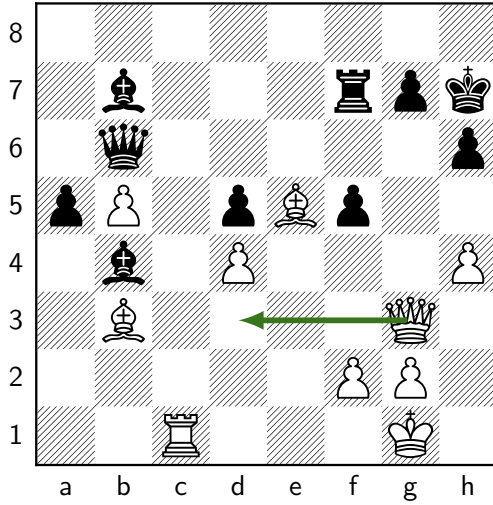
Figure 27: **Move 36, Black: Bg3**. (a) Our position; Melanie plays Bg3 (e5g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#1425` (ply 64) [structural distance $d=77$, QAP obj=96]: the master played Qa4; under μ our *Be4* plays the role of that game’s *Qc2* (a **cross-type** correspondence); cross-type roles: our *Nf6* as their *Rc1*, our *qb4* as their *Be1*; transferred move e5g3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#309` (ply 72) [structural distance $d=85$, QAP obj=74]: the master played Rxa4; under μ our *Be4* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Ra8* as their *Pf2*, our *Bc8* as their *Ph3*; transferred move e5g3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0020#2241` (ply 62) [structural distance $d=88$, QAP obj=19]: the master played Qxh4; under μ our *Be4* plays the role of that game’s *Qg3* (a **cross-type** correspondence); cross-type roles: our *Pg7* as their *Rd1*, our *bc1* as their *pb5*; transferred move e5g3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bg3 (summed QAP objective 216, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



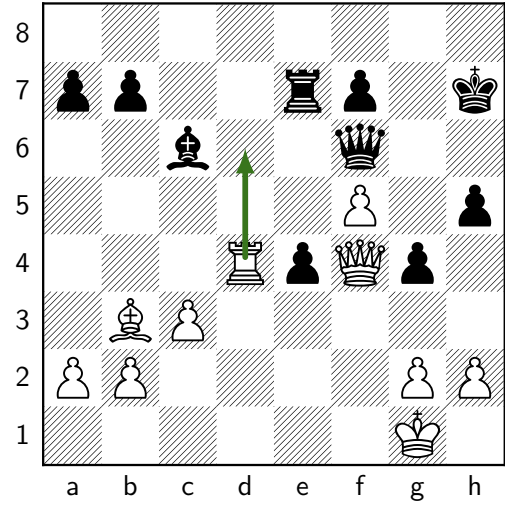
(a)



(b)

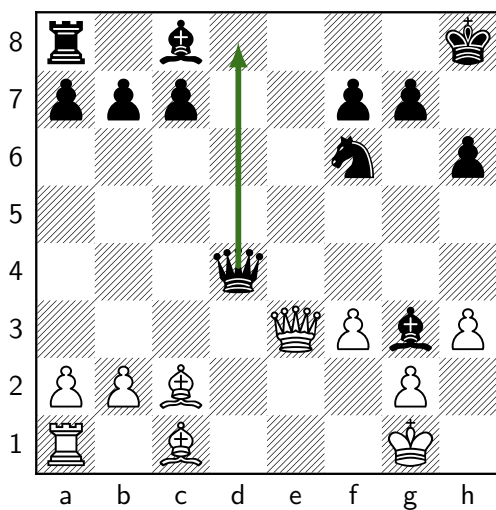


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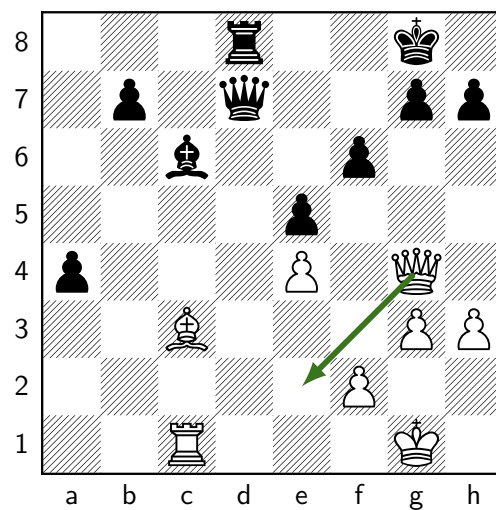


(d)

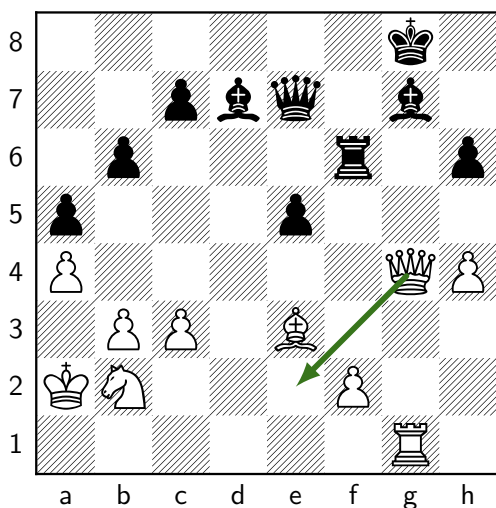
Figure 28: **Move 37, Black: Qd4+.** (a) Our position; Melanie plays Qd4+ (d5d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0023#1875` (ply 64) [structural distance $d=82$, QAP obj=71]: the master played Qd4; under μ our Qd4 plays the role of that game’s Qe3; cross-type roles: our Nf6 as their Re1, our Ra8 as their Pa2; transferred move d5d4 (**endorses this move**). (c) Retrieved from `twic1500#220` (ply 58) [structural distance $d=76$, QAP obj=36]: the master played Qd3; under μ our Qd4 plays the role of that game’s Qg3; cross-type roles: our pg2 as their Rc1, our Nf6 as their Pg2; transferred move d5d4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#1528` (ply 60) [structural distance $d=73$, QAP obj=12]: the master played Rd6; under μ our Qd4 plays the role of that game’s Rd4 (a **cross-type** correspondence); cross-type roles: our bc1 as their Pa2, our Ph6 as their Bb3; transferred move d5d4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd4+ (summed QAP objective 170, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



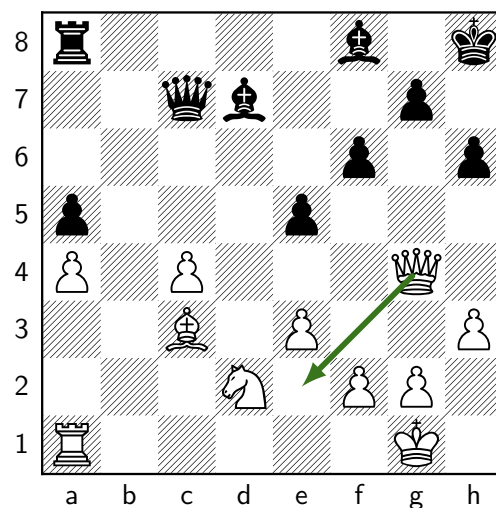
(a)



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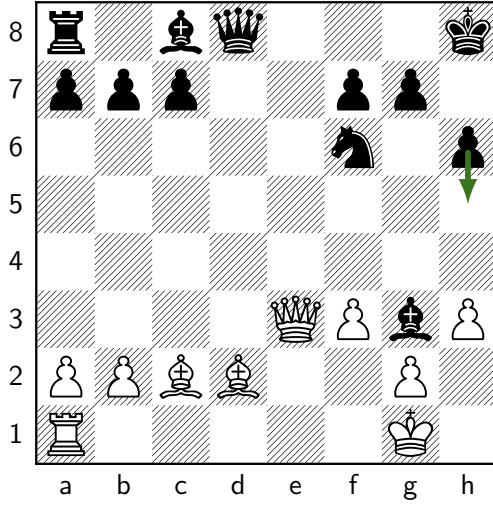


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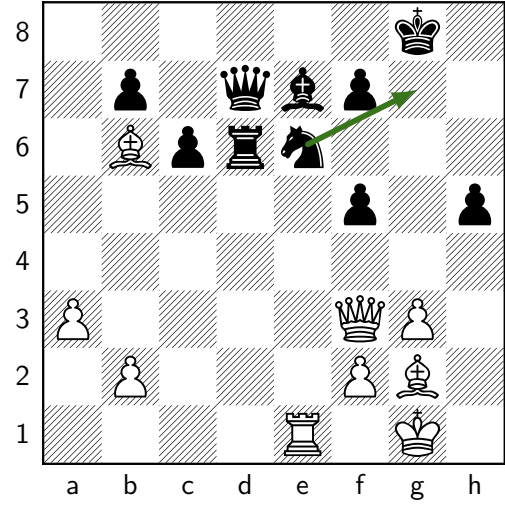


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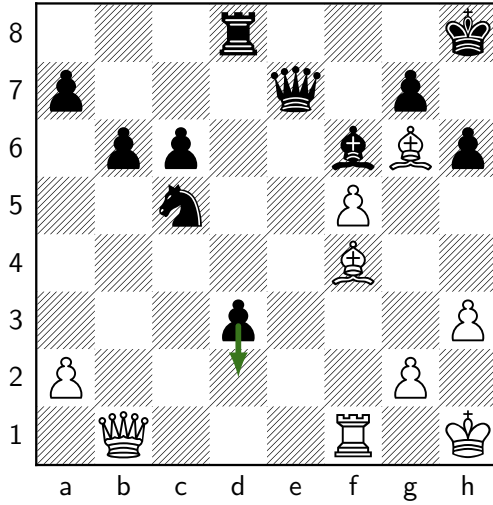
Figure 29: **Move 38, Black: Qd8**. (a) Our position; Melanie plays Qd8 (d4d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#2972` (ply 64) [structural distance $d=89$, QAP obj=25]: the master played Qe2; under μ our Qd5 plays the role of that game’s Qg4; cross-type roles: our Pf7 as their Rc1, our Nf6 as their Pg3; transferred move d4d8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#2105` (ply 64) [structural distance $d=89$, QAP obj=13]: the master played Qe2; under μ our Qd5 plays the role of that game’s Qg4; cross-type roles: our Pc7 as their Rg1, our Ph6 as their Nb2; transferred move d4d8 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#2614` (ply 86) [structural distance $d=88$, QAP obj=11]: the master played Qe2; under μ our Qd5 plays the role of that game’s Qg4; cross-type roles: our Bg3 as their Ra1, our Pc7 as their Nd2; transferred move d4d8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qd8 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



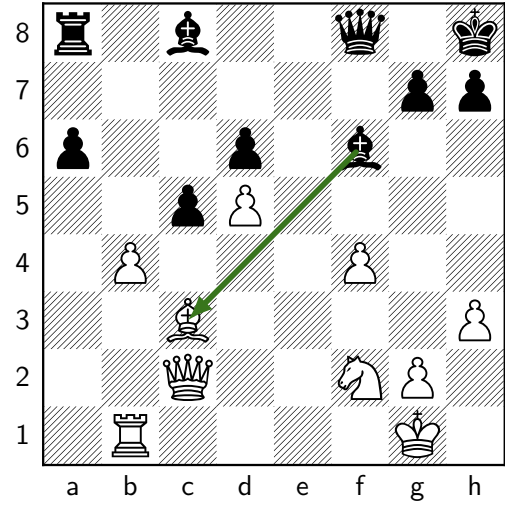
(a)



(b)

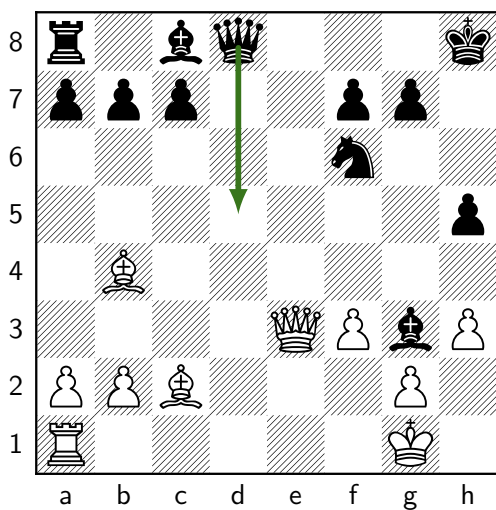


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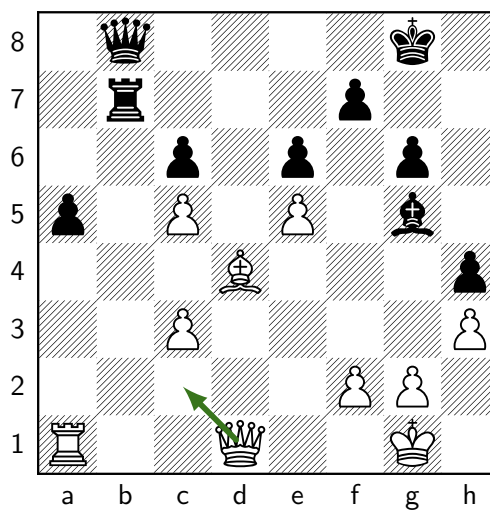


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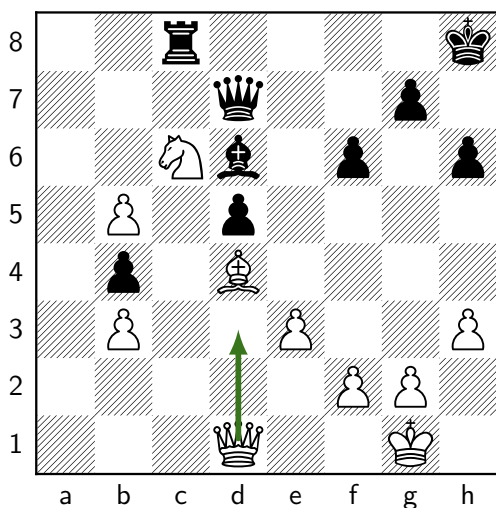
Figure 30: **Move 39, Black: h5**. (a) Our position; Melanie plays h5 (h6h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0022#3917 (ply 67) [structural distance $d=81$, QAP obj=49]: the master played Ng7; under μ our *Ph3* plays the role of that game’s *Ne3* (a **cross-type** correspondence); cross-type roles: our *Ra8* as their *Qd7*, our *ph3* as their *Be7*; transferred move h6h5 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0021#406 (ply 63) [structural distance $d=60$, QAP obj=43]: the master played d2; under μ our *Ph3* plays the role of that game’s *Pd6*; cross-type roles: our *Bg3* as their *Qe7*, our *Bc8* as their *Pg7*; transferred move h6h5 (**endorses this move**). (d) Retrieved from lichess_elite_2025-11_p0000#345 (ply 61) [structural distance $d=82$, QAP obj=20]: the master played Bxc3; under μ our *Ph3* plays the role of that game’s *Bf3* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *pf3* as their *Bc8*; transferred move h6h5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to h5 (summed QAP objective 144, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



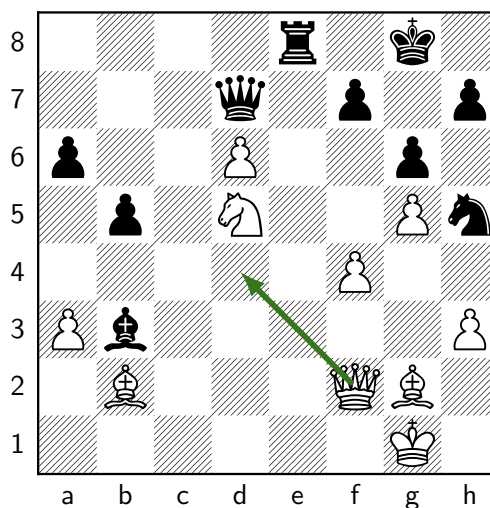
(a)



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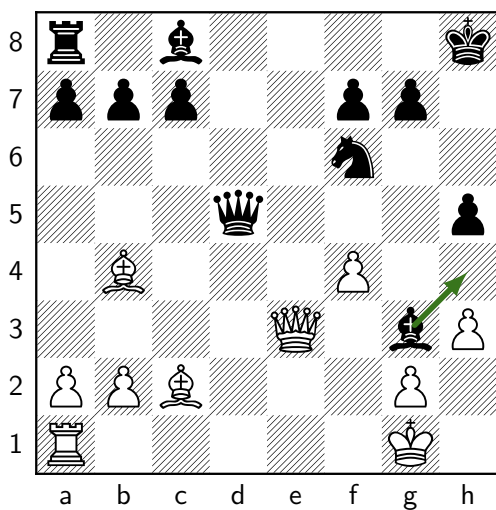


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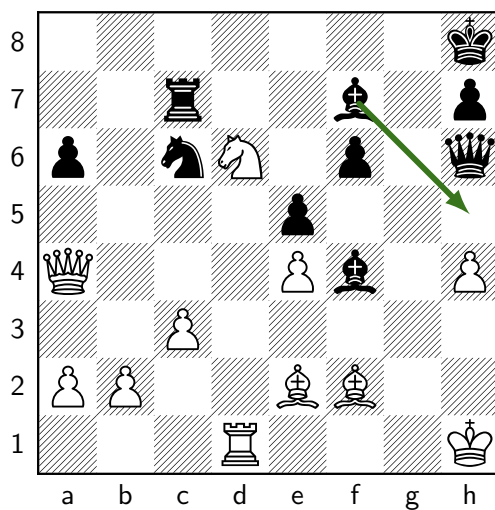


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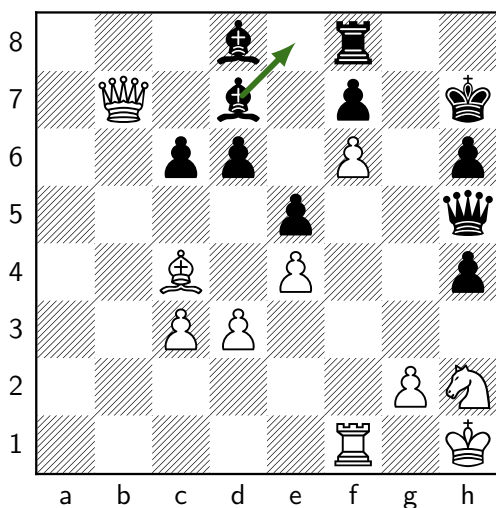
Figure 31: **Move 40, Black: Qd5.** (a) Our position; Melanie plays Qd5 (d8d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#309` (ply 68) [structural distance $d=78$, QAP obj=34]: the master played Qc2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Bc8 as their Ra1, our bc2 as their Pf2; transferred move d8d5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#804` (ply 76) [structural distance $d=83$, QAP obj=8]: the master played Qd3; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Bc8 as their Pe3, our Nf6 as their Ph3; transferred move d8d5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0018#1838` (ply 66) [structural distance $d=90$, QAP obj=4]: the master played Qd4; under μ our Qd1 plays the role of that game’s Qf2; cross-type roles: our Ra8 as their Bg2, our Bc8 as their Pa3; transferred move d8d5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd5 (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



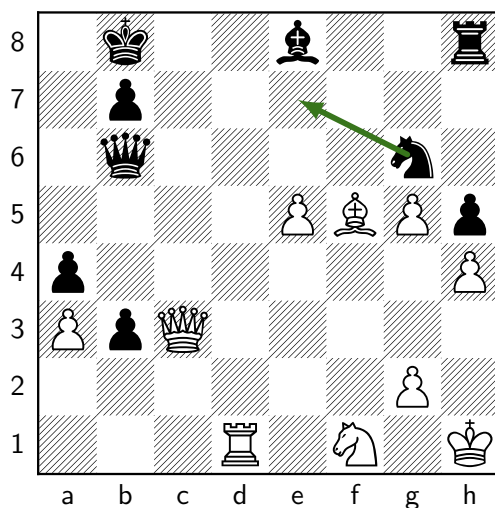
(a)



(b)

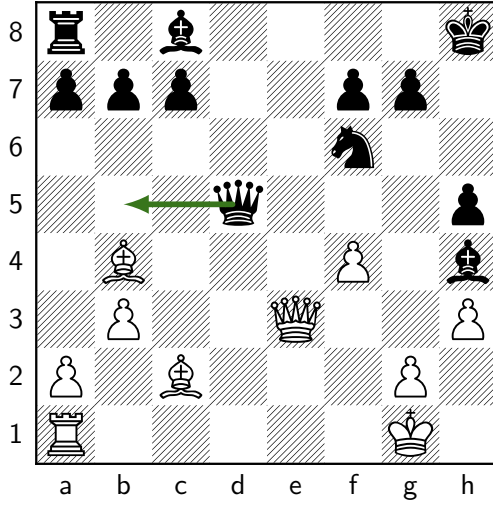


(c)

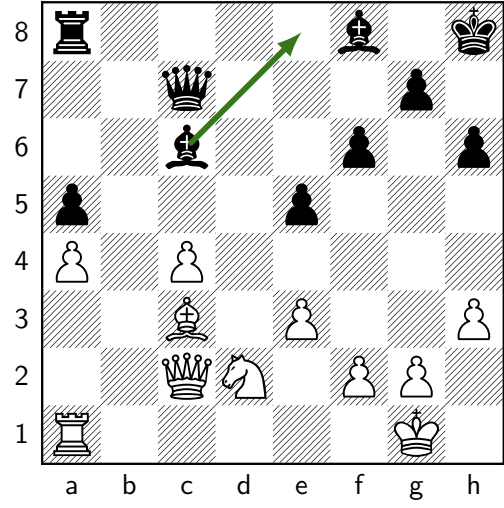


(d)

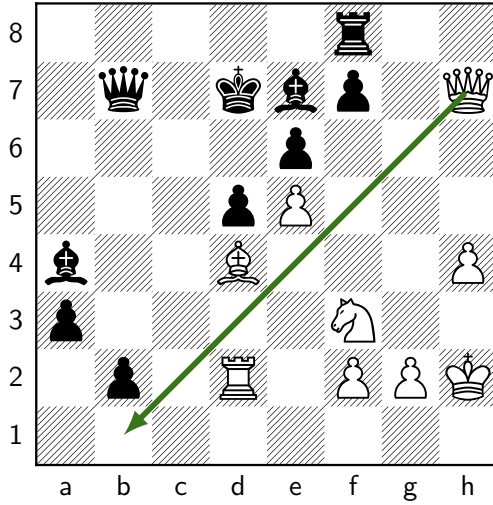
Figure 32: **Move 41, Black: Bh4 (g3h4)**. (a) Our position; Melanie plays Bh4 (g3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0020#3062 (ply 65) [structural distance $d=93$, QAP obj=37]: the master played Bh5; under μ our Bg6 plays the role of that game’s Bf2; cross-type roles: our Pa7 as their Rc7, our Nf6 as their Ph7; transferred move g3h4 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0010#1824 (ply 73) [structural distance $d=92$, QAP obj=33]: the master played Be8; under μ our Bg6 plays the role of that game’s Bd2; cross-type roles: our Qd5 as their Bd8, our Nf6 as their Rf8; transferred move g3h4 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0026#2187 (ply 71) [structural distance $d=81$, QAP obj=32]: the master played Ne7; under μ our Bg6 plays the role of that game’s Ng3 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Be8, our Pb7 as their Rh8; transferred move g3h4 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Bh4 (summed QAP objective 199, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



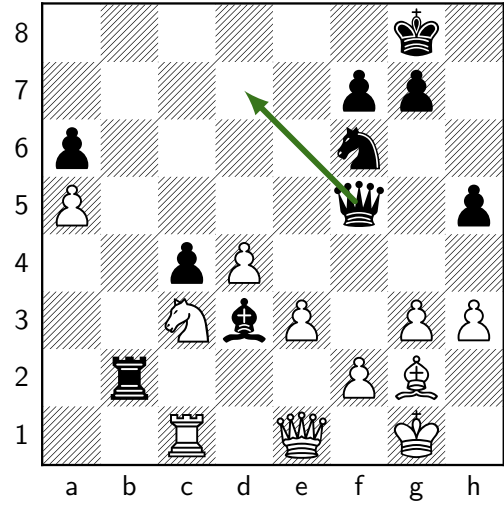
(a)



(b)

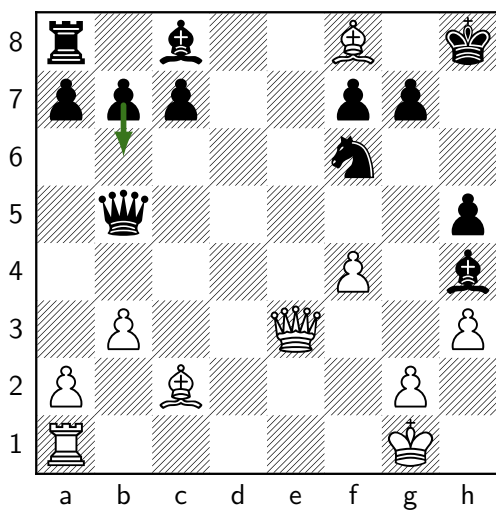


(c)

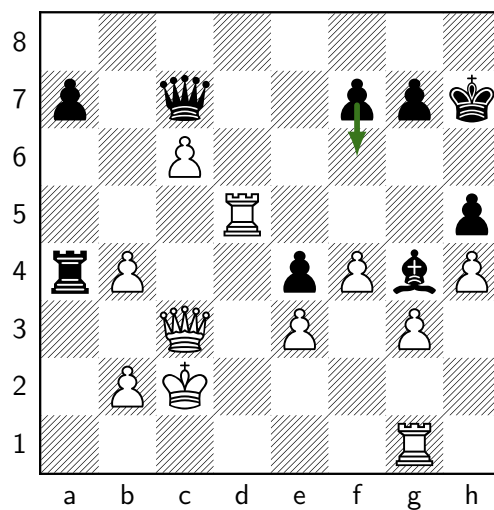


(d)

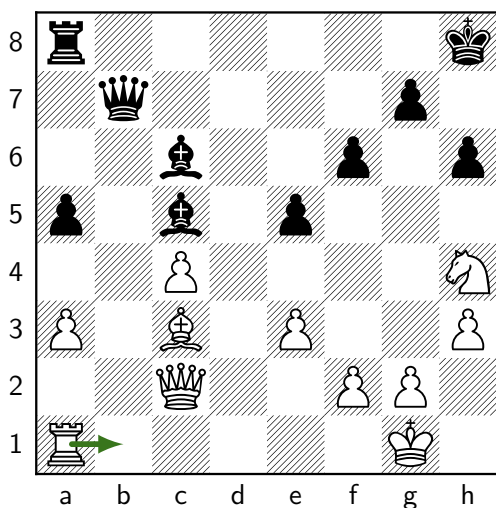
Figure 33: **Move 42, Black: Qb5**. (a) Our position; Melanie plays Qb5 (d5b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#2614` (ply 81) [structural distance $d=103$, QAP obj=60]: the master played Be8; under μ our Qd4 plays the role of that game’s Bc3 (a **cross-type** correspondence); cross-type roles: our Ph5 as their Ra8, our bc2 as their Pf6; transferred move d5b5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#1879` (ply 90) [structural distance $d=99$, QAP obj=57]: the master played Qb1; under μ our Qd4 plays the role of that game’s Qh7; cross-type roles: our Qd5 as their Rd2, our ra1 as their ba4; transferred move d5b5 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#2007` (ply 69) [structural distance $d=90$, QAP obj=52]: the master played Qd7; under μ our Qd4 plays the role of that game’s Qf4; cross-type roles: our Ra8 as their Pf7, our Bh4 as their Pa6; transferred move d5b5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qb5 (summed QAP objective 240, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



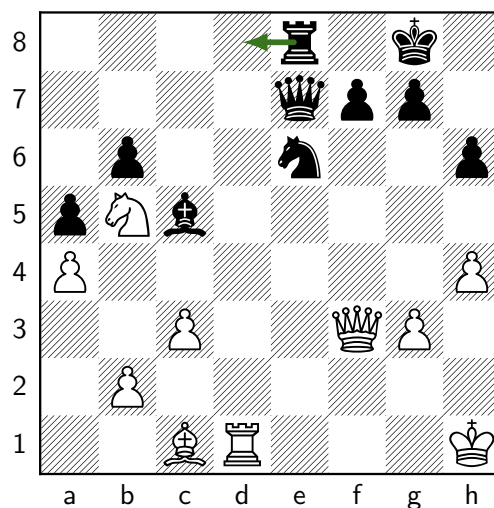
(a)



(b)

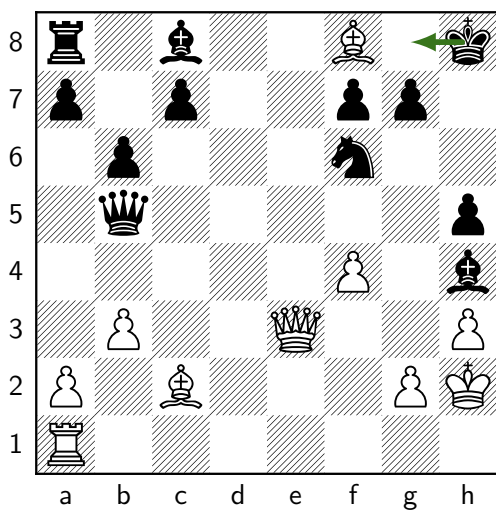


(c)

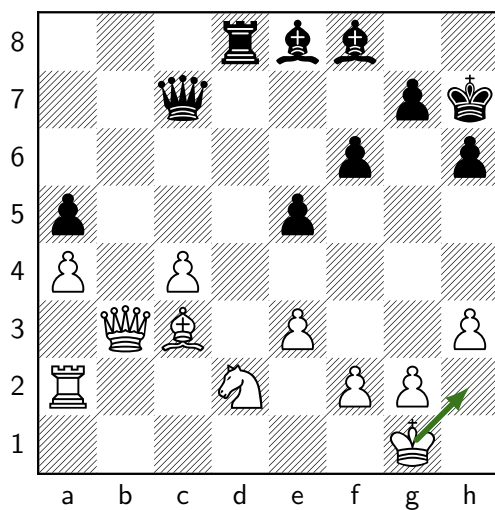


(d)

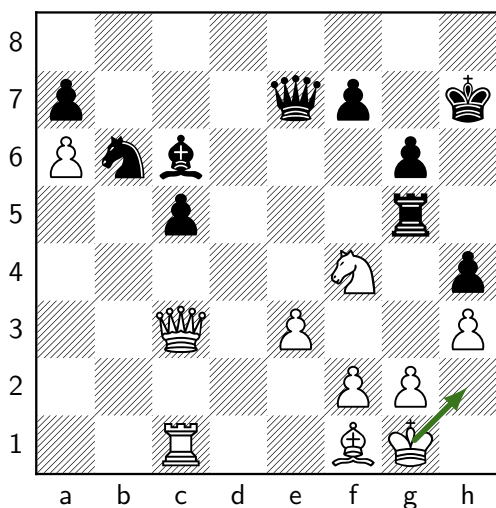
Figure 34: **Move 43, Black: b6**. (a) Our position; Melanie plays b6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#3983` (ply 81) [structural distance $d=100$, QAP obj=42]: the master played f6; under μ our *Pb2* plays the role of that game’s *Pf2*; cross-type roles: our *Nf6* as their *Pa7*, our *b7* as their *Qc7*; transferred move b7b6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#2614` (ply 68) [structural distance $d=84$, QAP obj=37]: the master played Rb1; under μ our *Pb2* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Qc2*, our *Qb5* as their *Bc3*; transferred move b7b6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0005#1445` (ply 69) [structural distance $d=95$, QAP obj=21]: the master played Rd8; under μ our *Pb2* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *Ph5* as their *Ne6*, our *Nf6* as their *Pa5*; transferred move b7b6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to b6 (summed QAP objective 180, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



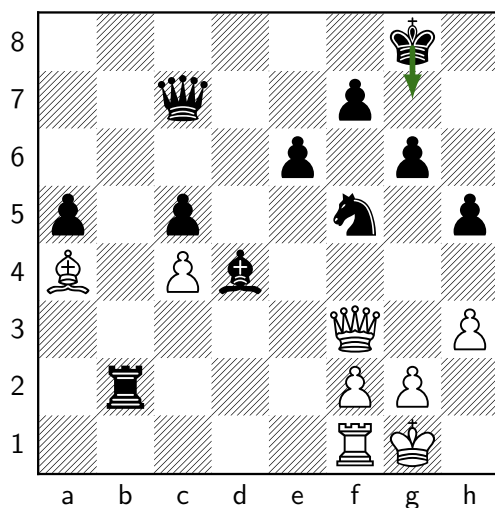
(a)



(b)

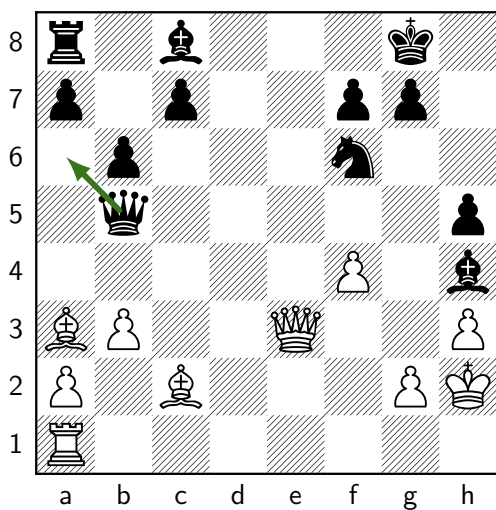


(c)

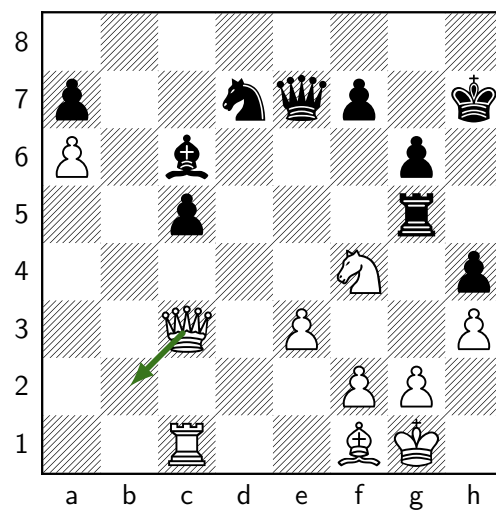


(d)

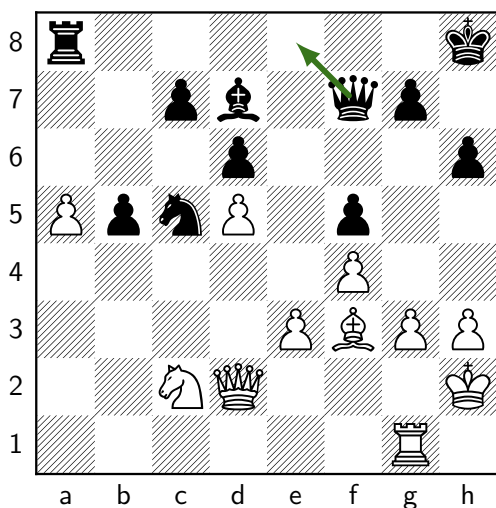
Figure 35: **Move 44, Black: Kg8.** (a) Our position; Melanie plays Kg8 (**h8g8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#2614` (ply 96) [structural distance $d=71$, QAP obj=35]: the master played Kh2; under μ our Kh1 plays the role of that game’s Kg1; cross-type roles: our Pg7 as their Nd2, our Nf6 as their Qb3; transferred move **h8g8 (endorses this move)**. (c) Retrieved from `lichess_elite_2024-09_p0024#821` (ply 84) [structural distance $d=94$, QAP obj=31]: the master played Kh2; under μ our Kh1 plays the role of that game’s Kg1; cross-type roles: our Pa7 as their Bf1, our bf8 as their Qc3; transferred move **h8g8 (endorses this move)**. (d) Retrieved from `lichess_elite_2024-09_p0022#1903` (ply 73) [structural distance $d=95$, QAP obj=22]: the master played Kg7; under μ our Kh1 plays the role of that game’s Kg1; cross-type roles: our Bc8 as their Qc7, our Nf6 as their Pe6; transferred move **h8g8 (endorses this move)**. Synthesis: 6 retrieved analogs transferred to Kg8 (summed QAP objective 154, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



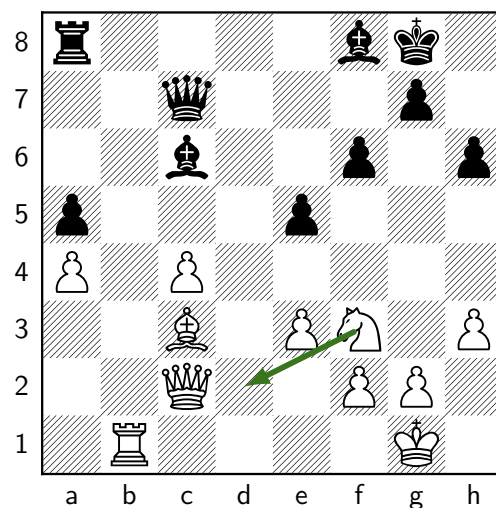
(a)



(b)

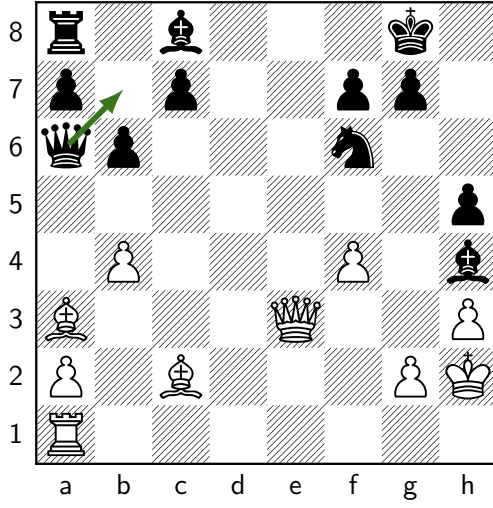


(c)

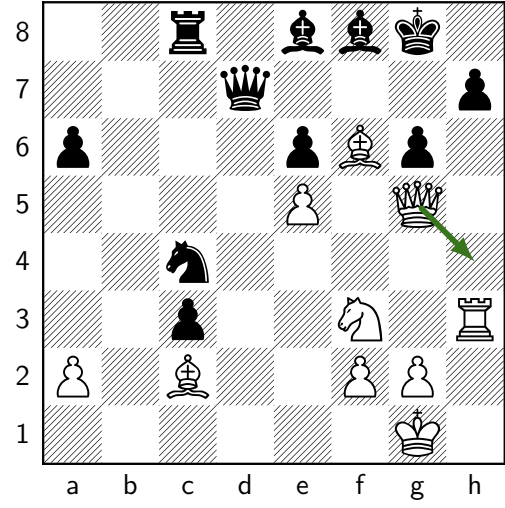


(d)

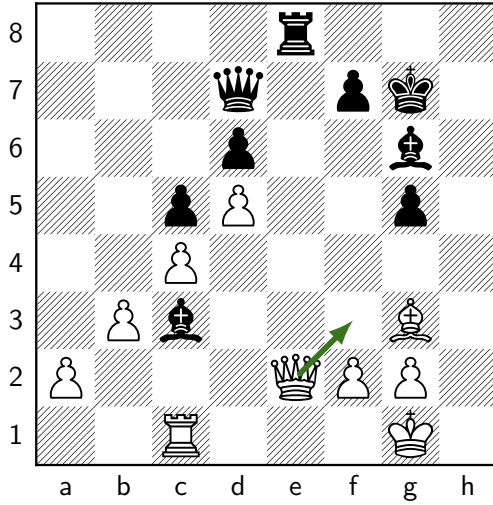
Figure 36: **Move 45, Black: Qa6**. (a) Our position; Melanie plays Qa6 (b5a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#821` (ply 72) [structural distance $d=95$, QAP obj=51]: the master played Qb2; under μ our Qb4 plays the role of that game’s Qc3; cross-type roles: our Ph5 as their Rc1, our ba3 as their Ph3; transferred move b5a6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#1452` (ply 77) [structural distance $d=96$, QAP obj=33]: the master played Qe8; under μ our Qb4 plays the role of that game’s Qf2; cross-type roles: our qe3 as their Ra8, our Ph5 as their Bd7; transferred move b5a6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#2614` (ply 74) [structural distance $d=90$, QAP obj=29]: the master played Nd2; under μ our Qb4 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Nf6 as their Rb1, our ba3 as their Qc2; transferred move b5a6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qa6 (summed QAP objective 119, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



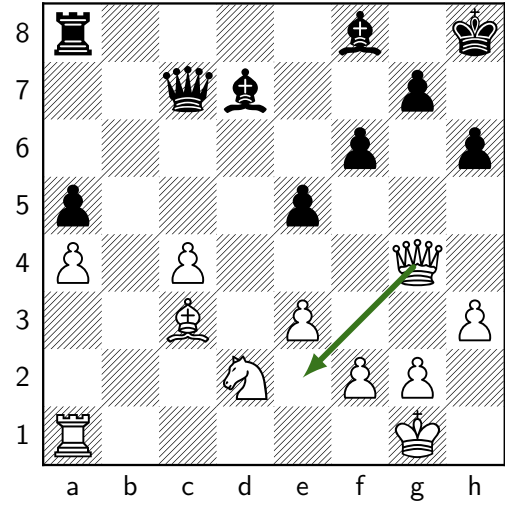
(a)



(b)

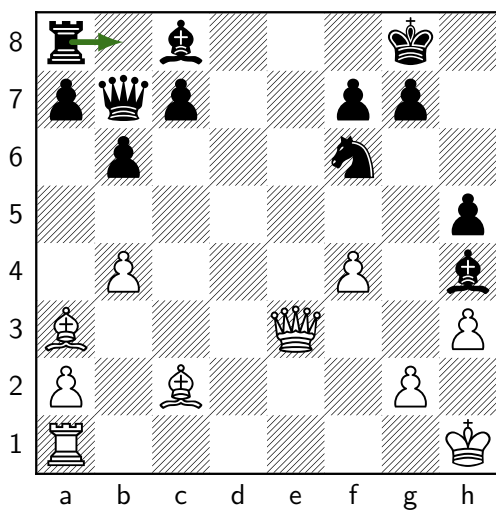


(c)

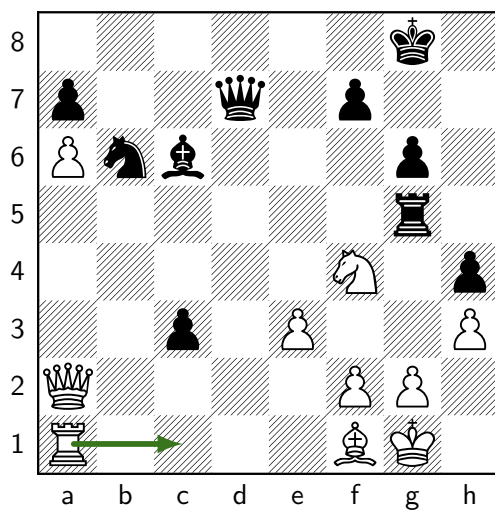


(d)

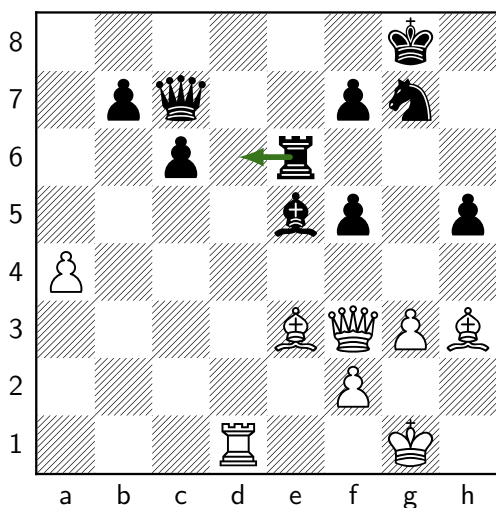
Figure 37: **Move 46, Black: Qb7.** (a) Our position; Melanie plays Qb7 (a6b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#2474` (ply 74) [structural distance $d=82$, QAP obj=33]: the master played Qh4; under μ our Qa3 plays the role of that game’s Qg5; cross-type roles: our Bh4 as their Pa2, our Qa6 as their Bc2; transferred move a6b7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0023#832` (ply 74) [structural distance $d=86$, QAP obj=19]: the master played Qf3; under μ our Qa3 plays the role of that game’s Qe2; cross-type roles: our Qa6 as their Rc1, our Nf6 as their Bg3; transferred move a6b7 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#2614` (ply 86) [structural distance $d=88$, QAP obj=14]: the master played Qe2; under μ our Qa3 plays the role of that game’s Qg4; cross-type roles: our Nf6 as their Ra1, our Pc7 as their Nd2; transferred move a6b7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qb7 (summed QAP objective 89, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



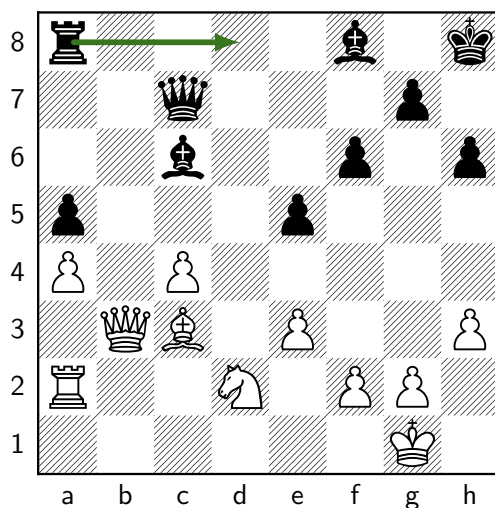
(a)



(b)

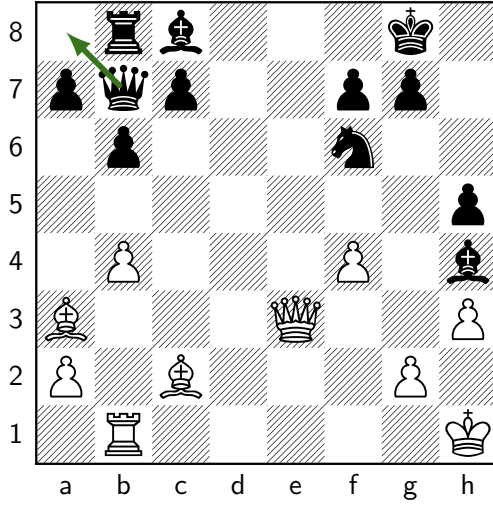


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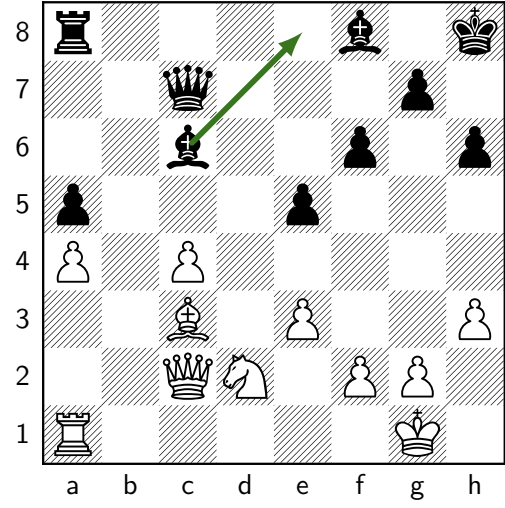


(d)

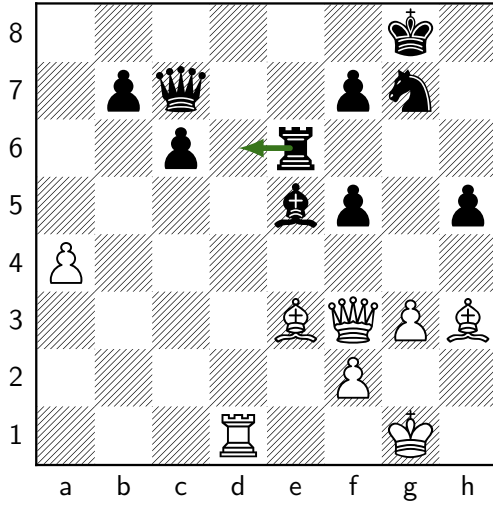
Figure 38: **Move 47, Black: Rb8.** (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#821` (ply 102) [structural distance $d=86$, QAP obj=67]: the master played Rc1; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *pf4* as their *Bf1*, our *ra1* as their *pc3*; transferred move a8b8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#3917` (ply 79) [structural distance $d=84$, QAP obj=54]: the master played Rd6; under μ our *Ra1* plays the role of that game’s *Re3*; cross-type roles: our *Pa7* as their *Ng7*, our *qe3* as their *bh3*; transferred move a8b8 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#2614` (ply 103) [structural distance $d=88$, QAP obj=39]: the master played Rd8; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *Bc8* as their *Qc7*, our *Qb7* as their *Bc6*; transferred move a8b8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rb8 (summed QAP objective 196, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



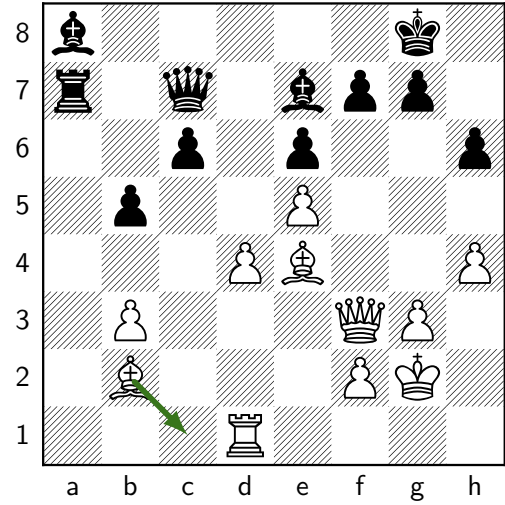
(a)



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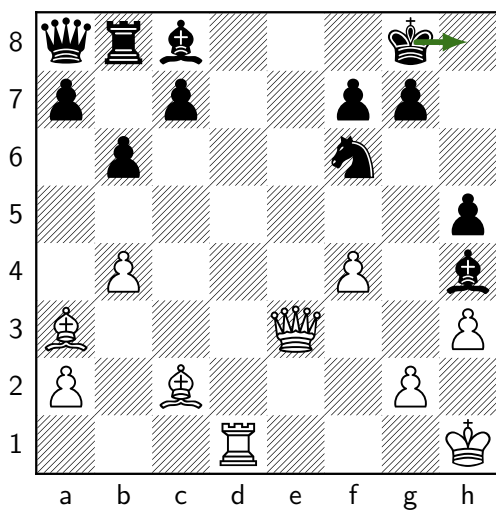


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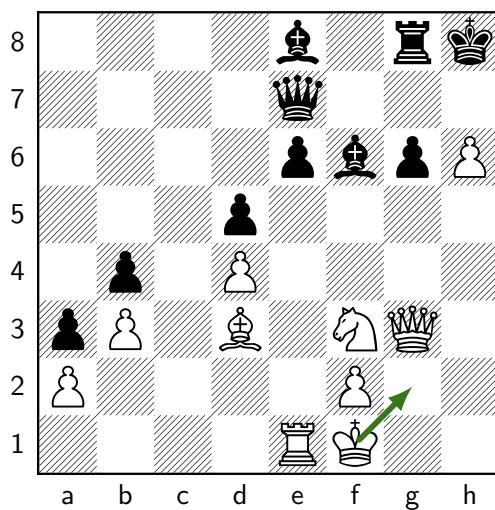


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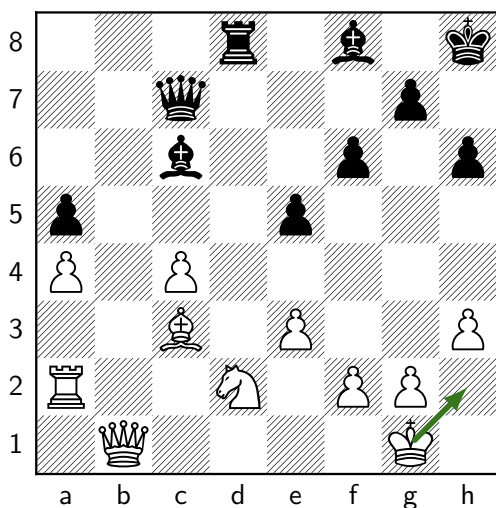
Figure 39: **Move 48, Black: Qa8.** (a) Our position; Melanie plays Qa8 (b7a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#2614` (ply 81) [structural distance $d=76$, QAP obj=44]: the master played Be8; under μ our Qb2 plays the role of that game’s Bc3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Qc7, our Nf6 as their Ph6; transferred move b7a8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#3917` (ply 79) [structural distance $d=88$, QAP obj=54]: the master played Rd6; under μ our Rb1 plays the role of that game’s Re3; cross-type roles: our Pa7 as their Ng7, our qe3 as their bh3; transferred move b8a8. (d) Retrieved from `lichess_elite_2024-09_p0022#2694` (ply 90) [structural distance $d=87$, QAP obj=41]: the master played Bc1; under μ our Bh5 plays the role of that game’s Bb2; cross-type roles: our Nf6 as their Rd1, our Rb8 as their Pf2; transferred move h4g5. Synthesis: 3 retrieved analogs transferred to Qa8 (summed QAP objective 129, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



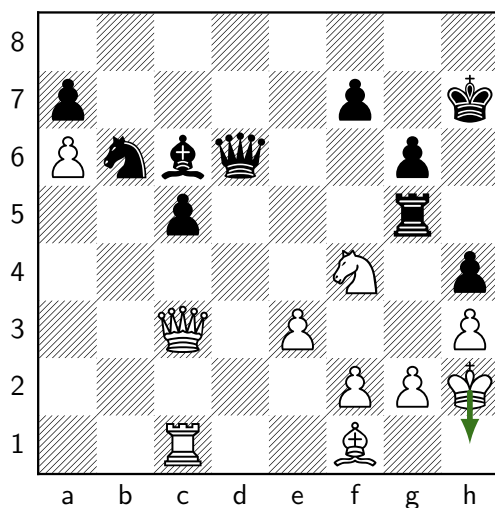
(a)



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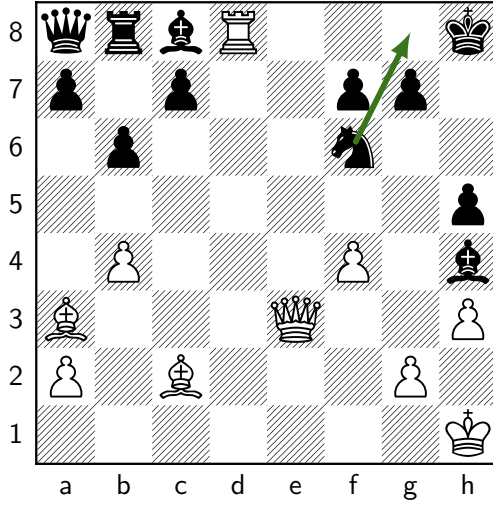


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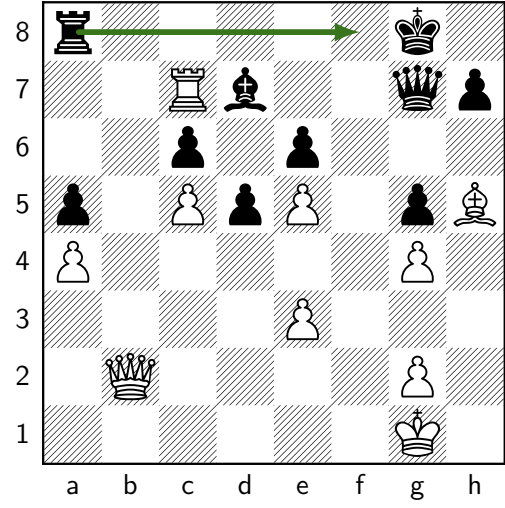


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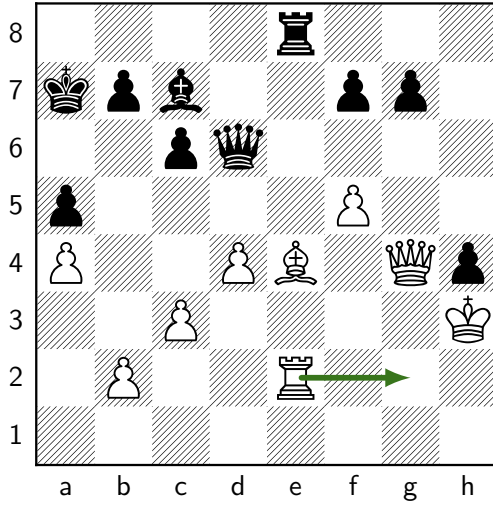
Figure 40: **Move 49, Black: Kh8**. (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#3603` (ply 82) [structural distance $d=92$, QAP obj=45]: the master played Kg2; under μ our Kg1 plays the role of that game’s Kf1; cross-type roles: our Qa8 as their Re1, our Bh4 as their pa3; transferred move g8h8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#2614` (ply 112) [structural distance $d=85$, QAP obj=42]: the master played Kh2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our rd1 as their Qb1, our pb4 as their Ra2; transferred move g8h8 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0024#821` (ply 86) [structural distance $d=82$, QAP obj=11]: the master played Kh1; under μ our Kg1 plays the role of that game’s Kh2; cross-type roles: our Bc8 as their Qc3, our bc2 as their Ph3; transferred move g8h8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kh8 (summed QAP objective 99, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



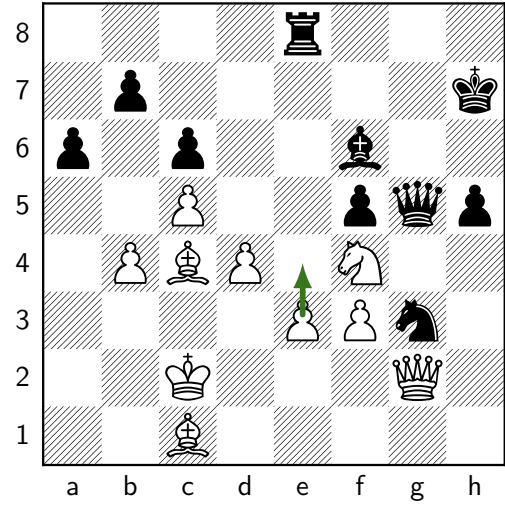
(a)



(b)

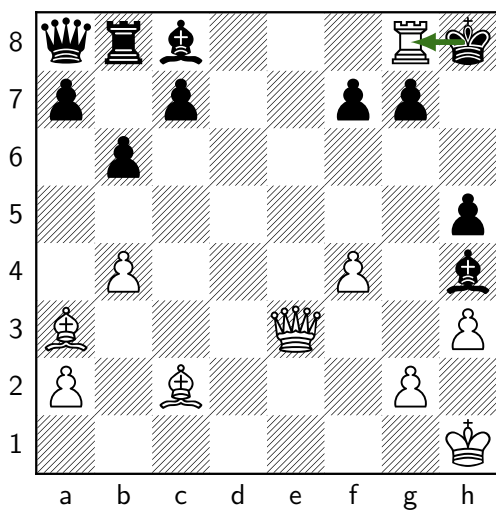


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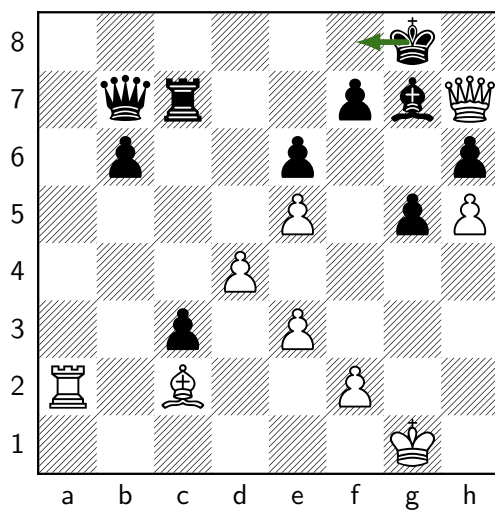


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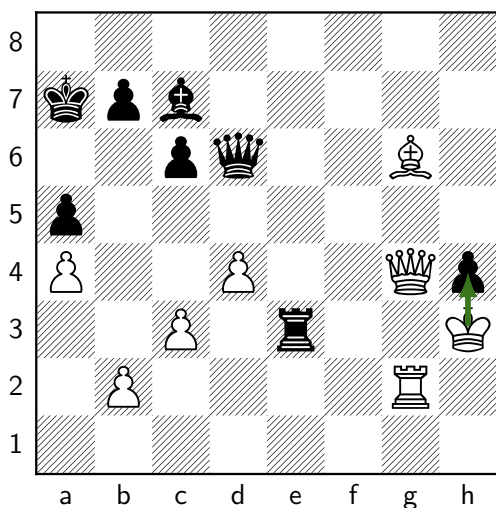
Figure 41: **Move 50, Black: Ng8**. (a) Our position; Melanie plays Ng8 (f6g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#66` (ply 91) [structural distance $d=96$, QAP obj=28]: the master played Rf8; under μ our Nf3 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Qg7, our qe3 as their bh5; transferred move f6g8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#1368` (ply 90) [structural distance $d=94$, QAP obj=26]: the master played Rg2; under μ our Nf3 plays the role of that game’s Re2 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Qg4, our bc2 as their ph4; transferred move f6g8 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#3946` (ply 90) [structural distance $d=100$, QAP obj=26]: the master played e4; under μ our Nf3 plays the role of that game’s Pe3 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Bc1, our rd8 as their Qg2; transferred move f6e8. Synthesis: 2 retrieved analogs transferred to Ng8 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



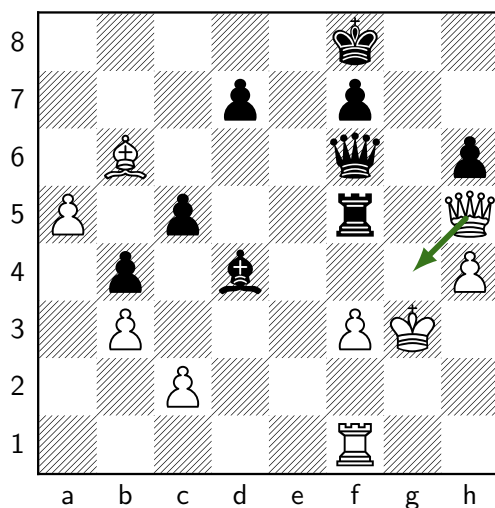
(a)



(b)



(c)



(d)

Figure 42: **Move 51, Black: Kxg8.** (a) Our position; Melanie plays Kxg8 (h8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#405` (ply 105) [structural distance $d=95$, QAP obj=78]: the master played Kf8; under μ our Kh1 plays the role of that game’s Kg1; cross-type roles: our Pg7 as their Bg7, our rg8 as their qh7; transferred move h8g8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#1368` (ply 96) [structural distance $d=96$, QAP obj=27]: the master played Kxh4; under μ our Kh1 plays the role of that game’s Kh3; cross-type roles: our Pg7 as their Rg2, our Rb8 as their Qg4; transferred move h8g8 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0025#3345` (ply 84) [structural distance $d=99$, QAP obj=21]: the master played Qg4; under μ our Kh1 plays the role of that game’s Qh5 (a **cross-type** correspondence); cross-type roles: our Bh4 as their Rf1, our Bc8 as their Pf3; transferred move h8g8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kxg8 (summed QAP objective 129, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.